

Overall no load voltage gain is

$$A_{V_T(NL)} = A_{V_1} \cdot A_{V_2} = (-1)(264.7) = -264.7$$

$$V_o = A_{V_T(NL)} V_i = (-264.7)(10 \text{ mV}) = -2.647 \text{ V}$$

Note that CE stage has a small voltage gain of unity and CB stage providing a larger voltage gain of 264.7.

(c) With $R_L = 10 \text{ k}\Omega$

$$A_{V_1} = -1 \quad (\text{same as before})$$

$$A_{V_2} = \frac{R_C \parallel R_L}{r_e} = \frac{(1.8 \text{ k}\Omega) \parallel (10 \text{ k}\Omega)}{6.8 \Omega} = 224.33$$

$$A_{V_T} = A_{V_1} \cdot A_{V_2} = -224.33$$

$$V_o = A_{V_T} V_i = (-224.33)(10 \text{ mV}) = -2.243 \text{ V}$$

(d) The input impedance is given by the input impedance of CE stage.

$$Z_i = \beta r_e = (200)(6.8 \Omega) = 1.36 \text{ k}\Omega$$

The output impedance is given by that of CB stage

$$Z_o = R_C = 1.8 \text{ k}\Omega$$

◆ 5.3 DARLINGTON CONNECTION

A Darlington connection is a very popular connection of two transistors for operation as one super beta transistor. The composite transistor acts as a single unit with a current gain equal to the product of the current gains of individual transistors. Figure 5.4 shows the Darlington connection which is also called the Darlington pair.

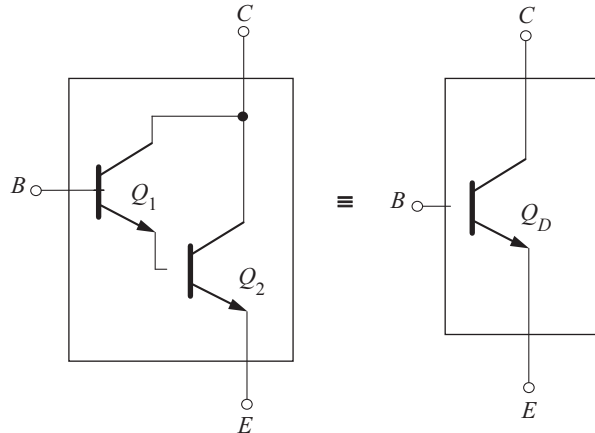


Fig. 5.4 Darlington connection

If β_1 and β_2 are the current gains of Q_1 and Q_2 respectively, the current gain of Darlington connection is

$$\beta_D = \beta_1 \beta_2 \quad (5.4)$$

If the two transistors are matched so that $\beta_1 = \beta_2 = \beta$, the Darlington connection provides a current gain of

$$\beta_D = \beta^2 \quad (5.5)$$

For instance if $\beta_1 = \beta_2 = 200$, the overall current gain is

$$\beta_D = (200)^2 = 40,000$$

A Darlington transistor connection acts as a single transistor with a large current gain, typically a few thousand.

Darlington connection is available as a single package containing two BJTs internally connected as a Darlington transistor. The device provides three terminals, base, emitter and collector for external connection. For instance the 2N 6725 is a Darlington transistor with a current gain of 25,000 and a collector current of 200 mA. As another example, the TIP 102 is a power Darlington transistor with a current gain of 1000 at a collector current of 3A.

◆ 5.4 DC BIAS OF DARLINGTON CIRCUIT

A Darlington circuit with biasing arrangement is shown in Fig. 5.5. The current gain of the Darlington transistor is β_D which is very high.

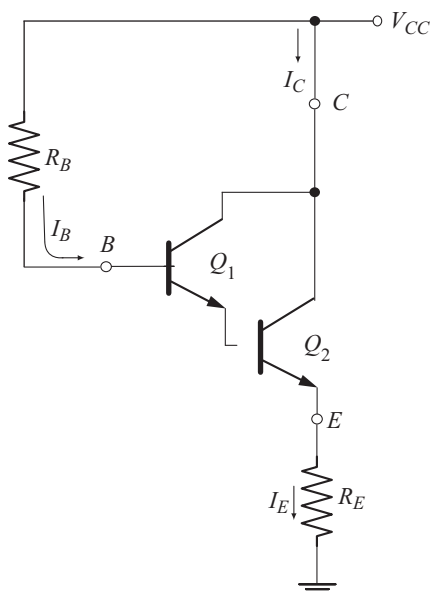


Fig. 5.5 DC bias of Darlington circuit

Applying KVL to the base-emitter circuit we have

$$V_{CC} = I_B R_B + V_{BE} + I_E R_E \quad (5.6)$$

$$I_E = (1 + \beta_D) I_B \quad (5.7)$$

Using Equation (5.7) in (5.6) and solving for I_B we have

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (1 + \beta_D) R_E} \quad (5.8)$$

It should be noted that, V_{BE} is the drop for two base-emitter junctions which is typically in the range 1.4 V to 1.8 V.

$$I_E = (1 + \beta_D) I_B \approx \beta_D I_B \quad [\text{Since } \beta_D \text{ is very high.}] \quad (5.9)$$

The dc voltages are

$$V_E = I_E R_E \quad (5.10)$$

$$V_B = V_{BE} + V_E \quad (5.11)$$

◆ 5.5 DARLINGTON EMITTER-FOLLOWER

Figure 5.6 shows the circuit of Darlington emitter-follower. The ac input signal V_i is coupled to the base of the Darlington transistor through the input coupling capacitor C_1 . The ac output voltage V_o is taken at the emitter through the output coupling capacitor C_2 .

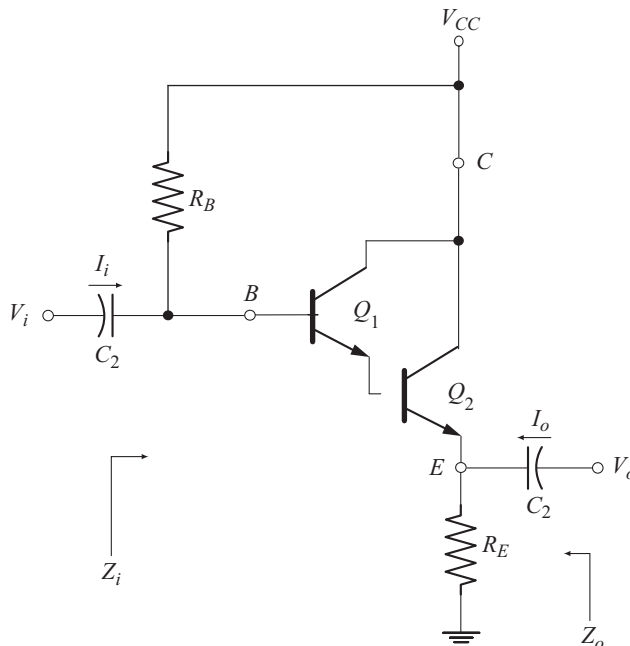


Fig. 5.6 Darlington emitter-follower

The ac equivalent circuit of Darlington emitter-follower is shown in Fig. 5.7. The Darlington transistor is replaced by

- (a) an input resistance r_i between base and emitter terminals
- (b) a controlled current source $\beta_D I_b$ between the collector and the emitter terminals.

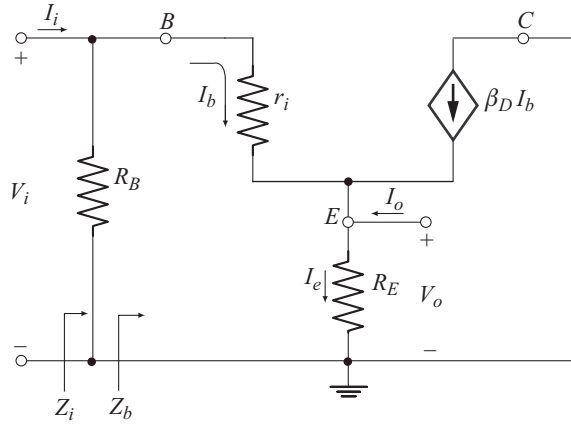


Fig. 5.7 AC equivalent circuit of Darlington emitter follower

AC Input Impedance (Z_i)

Applying KVL to the input circuit of Fig. 5.7 we have

$$V_i = I_b r_i + I_e R_E$$

Using $I_e = (1 + \beta_D) I_b$, we have

$$V_i = I_b r_i + (1 + \beta_D) R_E I_b$$

$$Z_b = \frac{V_i}{I_b} = r_i + (1 + \beta_D) R_E \quad (5.12)$$

Since β_D is very high,

$$Z_b \approx \beta_D R_E \quad (5.13)$$

$$Z_i = \frac{V_i}{I_i} = R_B \parallel Z_b \quad (5.14)$$

AC Current Gain (A_I)

$$A_I = \frac{I_o}{I_i} = \frac{I_o}{I_b} \cdot \frac{I_b}{I_i}$$

But

$$I_o = I_e$$

$\therefore$

$$A_I = \frac{I_e}{I_b} \cdot \frac{I_b}{I_i} \quad (5.15)$$

$$I_e = (1 + \beta_D) I_b \approx \beta_D I_b$$

$$\Rightarrow \frac{I_e}{I_b} = \beta_D \quad (5.16)$$

Applying KCL to the input circuit we have

$$I_i = \frac{V_i}{R_B} + I_b$$

Using $V_i = I_b Z_b$ we get

$$I_i = I_b \left[\frac{Z_b}{R_B} + 1 \right] = I_b \left[\frac{Z_b + R_B}{R_B} \right]$$

$$\Rightarrow \frac{I_b}{I_i} = \frac{R_B}{Z_b + R_B} \quad (5.17)$$

Using Equations (5.16) and (5.17) in Equation (5.15) we have

$$A_I = \beta_D \frac{R_B}{Z_b + R_B} \quad (5.18)$$

Substituting for Z_b from Equation (5.13), we get

$$A_I = \frac{\beta_D R_B}{R_B + \beta_D R_E} \quad (5.19)$$

AC Voltage Gain (A_V)

Refer the circuit of Fig. 5.7

$$\begin{aligned} V_o &= I_e R_E \\ &= [1 + \beta_D] I_b R_E \end{aligned}$$

From Equation (5.12) we have

$$\begin{aligned} V_i &= I_b [r_i + (1 + \beta_D) R_E] \\ \text{Now } A_V &= \frac{V_o}{V_i} = \frac{I_b [1 + \beta_D] R_E}{I_b [r_i + (1 + \beta_D) R_E]} \\ A_V &= \frac{[1 + \beta_D] R_E}{r_i + [1 + \beta_D] R_E} \quad (5.20) \end{aligned}$$

Since $(1 + \beta_D) R_E \gg r_i$

$$A_V \approx \frac{(1 + \beta_D) R_E}{[1 + \beta_D] R_E} = 1 \quad (5.21)$$

AC Output Impedance (Z_o)

To determine the ac output impedance we apply the following steps to the circuit of Fig. 5.7

- V_i is reduced to zero i.e. it is replaced by short circuit equivalent
- a voltage source V is connected between the output terminals.

The resulting circuit is shown in Fig. 5.8. Now the output impedance is given by

$$Z_o = \frac{V}{I} \quad (5.22)$$

where I is the current driven by the voltage source into the output terminals.

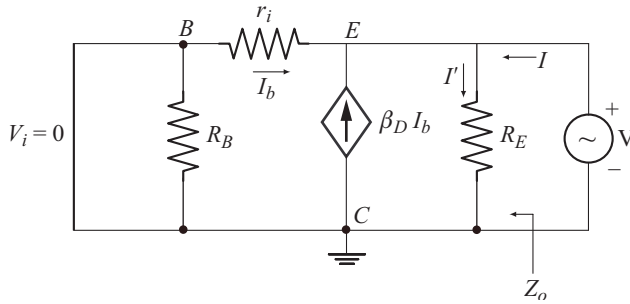


Fig. 5.8 Circuit to find Z_o .

The circuit of Fig. 5.8 is redrawn in Fig. 5.9.

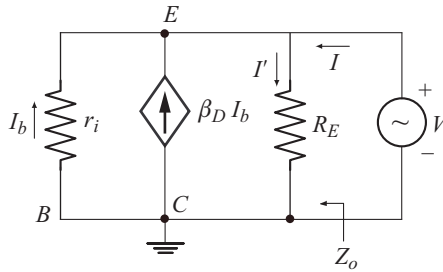


Fig. 5.9 Simplified circuit to find Z_o .

Applying KCL at the emitter node we have

$$I_b + \beta_D I_b - I' + I = 0 \quad (5.23)$$

$$\text{But } I_b = \frac{-V}{r_i} \quad \text{and} \quad I' = \frac{V}{R_E}$$

Using these relations in Equation (5.23) we get

$$-\frac{V}{r_i} + \beta_D \left(-\frac{V}{r_i} \right) - \frac{V}{R_E} + I = 0$$

$$\begin{aligned}
 \left[\frac{1}{r_i} + \frac{1}{R_E} + \frac{\beta_D}{r_i} \right] V &= I \\
 Z_o &= \frac{V}{I} = \frac{1}{\frac{1}{r_i} + \frac{1}{R_E} + \frac{\beta_D}{r_i}} \\
 Z_o &= \frac{1}{\frac{1}{r_i} + \frac{1}{R_E} + \frac{1}{(r_i/\beta_D)}} \quad (5.24)
 \end{aligned}$$

Z_o can be interpreted as the parallel combination of r_i , R_E and $\frac{r_i}{\beta_D}$

$$\therefore Z_o = r_i \parallel R_E \parallel \frac{r_i}{\beta_D} \quad (5.25)$$

Since $\frac{r_i}{\beta_D}$ is very much smaller than r_i and R_E we have

$$Z_o \approx \frac{r_i}{\beta_D} \quad (5.26)$$

5.5.1 Important Characteristics of Darlington Emitter-follower

The important characteristics of Darlington emitter-follower are:

- Very high current gain.
- Very high input impedance.
- Very low output impedance.
- Approximately unity voltage gain.
- Input and output voltages are in phase.

5.5.2 Applications of Darlington Emitter-follower

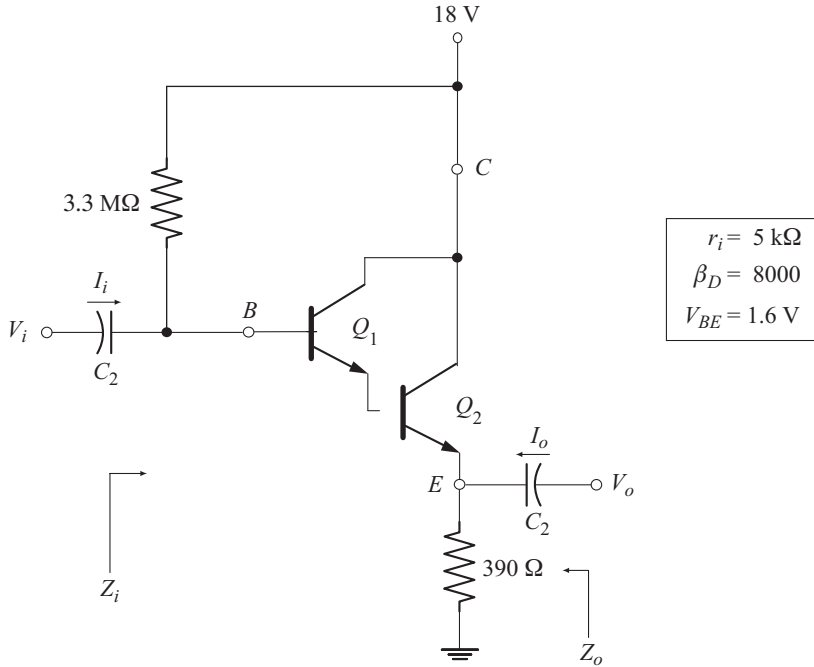
Darlington emitter-follower is a very popular configuration due to the attractive features listed above. Its important applications are:

- Buffer to connect a voltage source of high impedance to a load of low impedance.
- High current driver.
- Current amplifier.

Example 5.5

For the Darlington emitter-follower shown below:

- Calculate the dc bias voltages V_B , V_E , V_C and currents I_B and I_C .
- Calculate the input and output impedances.
- Determine the voltage and current gains.
- The ac output voltage for $V_i = 120$ mV.

**Solution**

$$(a) \quad I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta_D R_E}$$

$$= \frac{18 \text{ V} - 1.6 \text{ V}}{3.3 \text{ M}\Omega + (8000)(390 \Omega)} = 2.55 \mu\text{A}$$

$$I_E = I_{E2} \approx I_{C2} = \beta_D I_B$$

$$= (8000)(2.55 \mu\text{A}) = 20.4 \text{ mA}$$

$$V_E = I_E R_E = (20.4 \text{ mA})(390 \Omega) = 7.96 \text{ V}$$

$$V_B = V_{BE} + V_E = 1.6 \text{ V} + 7.96 \text{ V} = 9.56 \text{ V}$$

Since the collector is directly tied to V_{CC} , the collector voltage equals the dc supply voltage V_{CC} .

$$\therefore V_C = V_{CC} = 18 \text{ V}$$

$$(b) \quad Z_b = r_i + (1 + \beta_D) R_E$$

$$= 5 \text{ k}\Omega + (8001)(390 \Omega) = 3.13 \text{ M}\Omega$$

$$Z_i = R_B \parallel Z_b = 3.3 \text{ M}\Omega \parallel 3.13 \text{ M}\Omega = 1.6 \text{ M}\Omega$$

$$Z_o = r_i \parallel R_E \parallel \frac{r_i}{\beta_D}$$

$$= 5 \text{ k}\Omega \parallel 390 \text{ }\Omega \parallel \frac{5 \text{ k}\Omega}{8000} = 0.625 \text{ }\Omega$$

(c)

$$\begin{aligned} A_V &= \frac{R_E (1 + \beta_D)}{r_i + R_E (1 + \beta_D)} \\ &= \frac{(390 \text{ }\Omega)(8001)}{5 \text{ k}\Omega + (390 \text{ }\Omega)(8001)} = 0.998 \\ A_I &= \frac{\beta_D R_B}{R_B + \beta_D R_E} \\ &= \frac{(8000)(3.3 \text{ M}\Omega)}{3.3 \text{ M}\Omega + (8000)(390 \text{ }\Omega)} = 4112.15 \end{aligned}$$

(d)

$$\begin{aligned} V_o &= A_V V_i = (0.998) (120 \text{ mV}) \\ &= 119.76 \text{ mV} \end{aligned}$$

◆ 5.6 FEEDBACK PAIR

Figure 5.10 shows the feedback pair connection which is a two-transistor circuit that operates like the Darlington circuit. It is also called the complementary Darlington since it uses an npn and a pnp transistor. The collector current of Q_1 is the base current of Q_2 .

If the pnp transistor has a current gain of β_1 and the npn output transistor has a current gain of β_2 , the feedback pair acts like a single pnp transistor with a current gain of $\beta_1 \beta_2$.

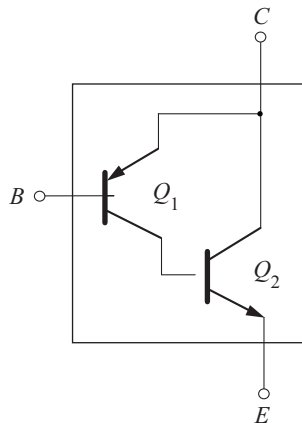


Fig. 5.10 Feedback pair connection

◆ 5.7 DC BIAS OF FEEDBACK PAIR

Figure 5.11 shows the biasing arrangement for feedback pair.

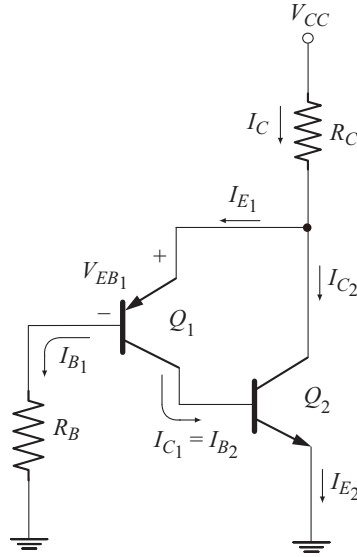


Fig . 5.11 DC biasing of feedback pair

Applying KVL to the base-emitter circuit of Q_1 we have

$$V_{CC} - I_C R_C - V_{EB1} - I_{B1} R_B = 0$$

Using $I_C = \beta_1 \beta_2 I_{B1}$, we have

$$V_{CC} - \beta_1 \beta_2 I_{B1} R_C - V_{EB1} - I_{B1} R_B = 0$$

$$I_{B1} = \frac{V_{CC} - V_{EB1}}{R_B + \beta_1 \beta_2 R_C} \quad (5.27)$$

The collector current of Q_1 is the base current of Q_2 .

$$\therefore I_{C1} = I_{B2} = \beta_1 I_{B1} \quad (5.28)$$

The collector current of Q_2 is

$$I_{C2} = \beta_2 I_{B2} = \beta_1 \beta_2 I_{B1} \approx I_{E2} \quad (5.29)$$

The current through R_C is

$$\begin{aligned} I_C &= I_{E1} + I_{C2} \\ &\approx I_{C1} + I_{C2} \\ &\approx I_{C2} \end{aligned} \quad (5.30)$$

Since

$$I_{C2} \gg I_{C1}$$

◆ 5.8 AC OPERATION OF FEEDBACK PAIR

The ac input signal V_i is coupled to the base of Q_1 through the input coupling capacitor C_1 as shown in Fig. 5.12. The ac output signal V_o is taken at the collector of Q_2 through the output coupling capacitor C_2 .

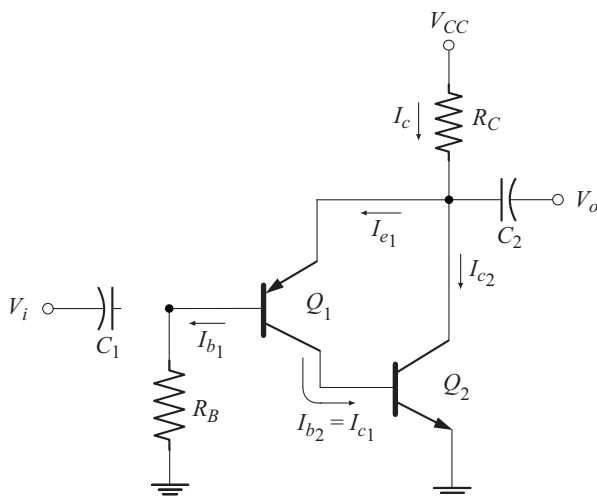


Fig. 5.12 AC operation of feedback pair

Figure 5.13 shows the ac equivalent circuit of feedback pair. For the ease of analysis the ac equivalent circuit is redrawn as shown in Fig. 5.14. It is important to note that, same ac model is used for both *pnp* and *npn* transistors. The actual current directions are investigated when ever necessary.

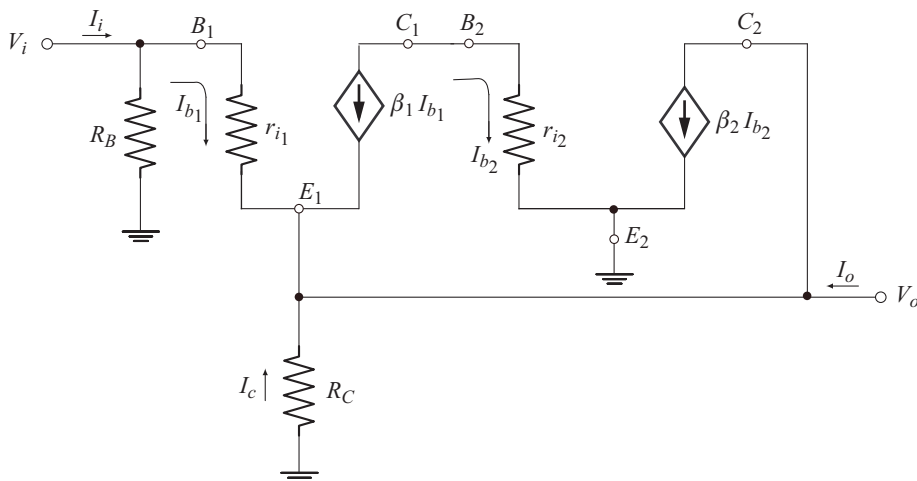


Fig. 5.13 AC equivalent circuit of feedback pair

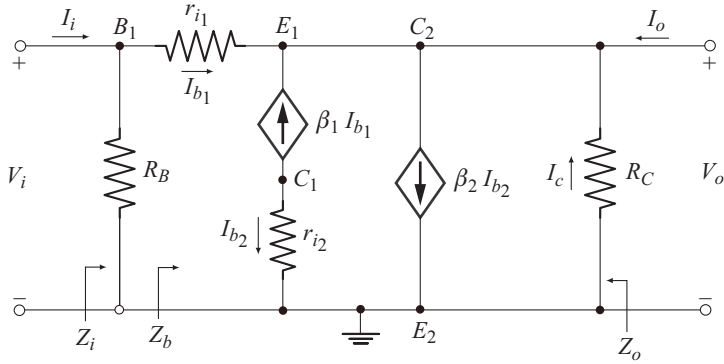


Fig. 5.14 AC equivalent circuit redrawn

AC Input Impedance (Z_i)

Writing KVL to the outer path of Fig. 5.14 we have

$$V_i - I_{b1} r_{i1} + I_c R_C = 0 \quad (5.31)$$

From the circuit of Fig. 5.12

$$\begin{aligned} I_c &= I_{e1} + I_{c2} \\ I_{c2} &= \beta_2 I_{b2} \quad \text{and} \quad I_{e1} = (1 + \beta_1) I_{b1} \approx \beta_1 I_{b1} \\ \text{Now} \quad I_c &= \beta_2 I_{b2} + \beta_1 I_{b1} \\ \text{But} \quad \beta_2 I_{b2} &\gg \beta_1 I_{b1} \\ \therefore \quad I_c &\approx \beta_2 I_{b2} \\ \text{But} \quad I_{b2} &= I_{c1} = \beta_1 I_{b1} \\ \text{So that} \quad I_c &= \beta_1 \beta_2 I_{b1} \end{aligned}$$

Since I_c and I_{b1} are in opposite directions, it is appropriate to write

$$I_c = -\beta_1 \beta_2 I_{b1} \quad (5.32)$$

Using this relation in Equation (5.31) we have

$$\begin{aligned} V_i &= I_{b1} r_{i1} + I_{b1} \beta_1 \beta_2 R_C \\ \therefore \quad Z_b &= \frac{V_i}{I_{b1}} = r_{i1} + \beta_1 \beta_2 R_C \end{aligned} \quad (5.33)$$

$$Z_i = R_B \parallel Z_b \quad (5.34)$$

AC Current Gain (A_I)

$$A_I = \frac{I_o}{I_i} = \frac{I_o}{I_{b1}} \cdot \frac{I_{b1}}{I_i} \quad (5.35)$$

But $I_o = -I_c$

∴ From Equation (5.32),
$$\frac{I_o}{I_{b_1}} = \frac{-I_c}{I_{b_1}} = \beta_1 \beta_2 \quad (5.36)$$

Applying KCL at the node B_1 of Fig. 5.14 we have

$$I_i = I_{b_1} + \frac{V_i}{R_B}$$

Using $V_i = I_{b_1} Z_b$ we have

$$\begin{aligned} I_i &= I_{b_1} \left[1 + \frac{Z_b}{R_B} \right] \\ \frac{I_i}{I_{b_1}} &= \frac{R_B + Z_b}{R_B} \\ \text{or} \quad \frac{I_{b_1}}{I_i} &= \frac{R_B}{R_B + Z_b} \end{aligned} \quad (5.37)$$

Using Equations (5.36) and (5.37) in Equation (5.35) we have

$$A_I = \frac{I_o}{I_i} = \beta_1 \beta_2 \frac{R_B}{R_B + Z_b} \quad (5.38)$$

AC Voltage Gain (A_v)

$$V_o = I_c R_c$$

Substituting for I_c from Equation (5.32) we have

$$V_o = \beta_1 \beta_2 I_{b_1} R_c \quad [\text{Taking only magnitude}] \quad (5.39).$$

From Equation (5.31) we have

$$\begin{aligned} V_i - I_{b_1} r_{i_1} - V_o &= 0 \\ \Rightarrow I_{b_1} &= \frac{V_i - V_o}{r_{i_1}} \end{aligned}$$

Using this relation in Equation (5.39) we get

$$\begin{aligned} V_o &= \beta_1 \beta_2 R_c \left[\frac{V_i - V_o}{r_{i_1}} \right] \\ V_o &= \frac{\beta_1 \beta_2 R_c}{r_{i_1}} V_i - \frac{\beta_1 \beta_2 R_c}{r_{i_1}} V_o \\ V_o \left[1 + \frac{\beta_1 \beta_2 R_c}{r_{i_1}} \right] &= V_i \left[\frac{\beta_1 \beta_2 R_c}{r_{i_1}} \right] \end{aligned}$$

$$A_V = \frac{V_o}{V_i} = \left[\frac{\beta_1 \beta_2 R_C}{r_{i1} + \beta_1 \beta_2 R_C} \right] \quad (5.40)$$

AC Output Impedance (Z_o)

The circuit to find Z_o is shown in Fig. 5.15, which is obtained by reducing V_i to zero and connecting an ac source of voltage V between the output terminals in the circuit of Fig. 5.14.

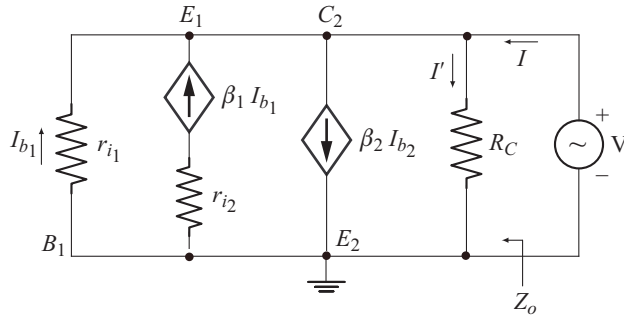


Fig. 5.15 Circuit to find Z_o .

Summing all the currents at the node C_2 , we have

$$I_{b1} + \beta_1 I_{b1} - \beta_2 I_{b2} - I' + I = 0 \quad (5.41)$$

$$I_{b1} = -\frac{V}{r_{i1}} \quad \text{and} \quad I' = \frac{V}{R_C}$$

$$I_{b2} = I_{c1} = \beta_1 I_{b1}$$

I_{b1} is the base current of *pnp* transistor and I_{b2} is that of *nnp* transistor. Thus taking direction into consideration we have

$$I_{b2} = -\beta_1 I_{b1} = -\beta_1 \left[\frac{-V}{r_{i1}} \right] = \frac{\beta_1 V}{r_{i1}}$$

Using these relations in Equation (5.41) we have

$$\frac{-V}{r_{i1}} - \beta_1 \frac{V}{r_{i1}} - \beta_1 \beta_2 \frac{V}{r_{i1}} - \frac{V}{R_C} + I = 0$$

$$V \left[\frac{1}{r_{i1}} + \frac{\beta_1}{r_{i1}} + \frac{\beta_1 \beta_2}{r_{i1}} + \frac{1}{R_C} \right] = I$$

$$Z_o = \frac{V}{I} = \frac{1}{\frac{1}{r_{i1}} + \frac{\beta_1}{r_{i1}} + \frac{\beta_1 \beta_2}{r_{i1}} + \frac{1}{R_C}}$$

$$\text{or} \quad Z_o = \frac{1}{\frac{1}{r_{i_1}} + \frac{1}{(r_{i_1}/\beta_1)} + \frac{1}{(r_{i_1}/\beta_1\beta_2)} + \frac{1}{R_C}} \quad (5.42)$$

From Equation (5.42), Z_o can be interpreted as the parallel combination of r_{i_1} , $\frac{r_{i_1}}{\beta_1}$, $\frac{r_{i_1}}{\beta_1\beta_2}$ and R_C .

$$\text{i.e.} \quad Z_o = r_{i_1} \parallel \frac{r_{i_1}}{\beta_1} \parallel \frac{r_{i_1}}{\beta_1\beta_2} \parallel R_C \quad (5.43)$$

Since $\frac{r_{i_1}}{\beta_1\beta_2}$ is the smallest of all, we can take

$$Z_o \approx \frac{r_{i_1}}{\beta_1\beta_2} \quad (5.44)$$

Example 5.6

For the feedback pair of Fig. 5.12 the following data are available:

$$\begin{aligned} R_B &= 2 \text{ M}\Omega & R_C &= 100 \text{ }\Omega & \beta_1 &= 140 & \beta_2 &= 180 \\ V_{CC} &= 18 \text{ V} & V_i &= 120 \text{ mV} & r_{i_1} &= 4 \text{ k}\Omega \end{aligned}$$

- Calculate the dc bias currents and Voltages.
- Calculate Z_i and Z_o .
- Calculate A_v and A_i .
- Determine V_o .

Solution

$$\begin{aligned} \text{(a)} \quad I_{B_1} &= \frac{V_{CC} - V_{EB_1}}{R_B + \beta_1\beta_2 R_C} \\ &= \frac{18 \text{ V} - 0.7 \text{ V}}{2 \text{ M}\Omega + (140)(180)(100 \text{ }\Omega)} = 3.83 \text{ }\mu\text{A} \\ I_{B_2} &= I_{C_1} = \beta_1 I_{B_1} = (140)(3.83 \text{ }\mu\text{A}) = 0.536 \text{ mA} \\ I_{C_2} &= \beta_2 I_{B_2} = (180)(0.536 \text{ mA}) = 96.48 \text{ mA} \\ I_C &= I_{E_1} + I_{C_2} \approx I_{C_1} + I_{C_2} \\ &= 0.536 \text{ mA} + 96.48 \text{ mA} = 97.01 \text{ mA} \\ V_{CC} &= I_C R_C + V_{CE_2} \\ V_{CE_2} &= V_{o(dc)} = V_{CC} - I_C R_C \\ &= 18 \text{ V} - (97.01 \text{ mA})(100 \text{ }\Omega) \\ &= 8.29 \text{ V} \quad [\text{DC output voltage}] \end{aligned}$$

$$\begin{aligned}
 \text{Also } V_{CC} &= I_C R_C - V_{EB_1} + V_{B_1} \\
 V_{B_1} &= V_{CC} - I_C R_C - V_{EB_1} \\
 \text{or } V_{B_1} &= V_{i(dc)} = V_{o(dc)} - V_{EB_1} = 8.29 \text{ V} - 0.7 \text{ V} \\
 &= 7.59 \text{ V} \quad [\text{DC input voltage}]
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad Z_b &= r_{i_1} + \beta_1 \beta_2 R_C \\
 &= 4 \text{ k}\Omega + (140)(180)(100 \Omega) = 2.524 \text{ M}\Omega \\
 Z_i &= R_B \parallel Z_b = 2 \text{ M}\Omega \parallel 2.524 \text{ M}\Omega = 1.12 \text{ M}\Omega \\
 Z_o &= r_{i_1} \parallel \frac{r_{i_1}}{\beta_1} \parallel \frac{r_{i_1}}{\beta_1 \beta_2} \parallel R_C \\
 &= 4 \text{ k}\Omega \parallel \frac{4 \text{ k}\Omega}{140} \parallel \frac{4 \text{ k}\Omega}{(140)(180)} \parallel 100 \Omega = 0.158 \Omega
 \end{aligned}$$

$$\begin{aligned}
 (c) \quad A_V &= \frac{\beta_1 \beta_2 R_C}{\beta_1 \beta_2 R_C + r_{i_1}} \\
 &= \frac{(140)(180)(100 \Omega)}{(140)(180)(100 \Omega) + 4 \text{ k}\Omega} = 0.9984
 \end{aligned}$$

$$\begin{aligned}
 A_I &= \beta_1 \beta_2 \frac{R_B}{R_B + Z_b} \\
 &= (140)(180) \frac{2 \text{ M}\Omega}{2 \text{ M}\Omega + 2.524 \text{ M}\Omega} \\
 &= 11.85 \times 10^3
 \end{aligned}$$

$$(d) \quad V_o = A_V V_i (0.9984)(120 \text{ mV}) = 119.81 \text{ mV}$$

◆ 5.9 CURRENT MIRROR CIRCUIT

Integrated circuit amplifiers are biased using current mirror circuits, which provide a constant current. The constant current is obtained from an output circuit, which is the reflection or mirror of a constant current developed on one side of the circuit. Figure 5.16 shows the circuit of a current mirror.

The circuit consists of two matched or identical transistors, Q_1 and Q_2 , operating at the same temperature, with their base and emitter terminals tied together. The base emitter voltage is therefore the same in two transistors.

The transistor Q_1 is connected as a diode by shorting its collector to base. When the supply Voltage V_{CC} is applied, the base-emitter junction of Q_1 is forward biased and a reference current I_X is established. We can think of V_{BE} as being the result of I_X . The same V_{BE} is applied to the base emitter junction of Q_2 . Since the two transistors are identical, the collector current of Q_2 will be the mirror of the reference current I_X .

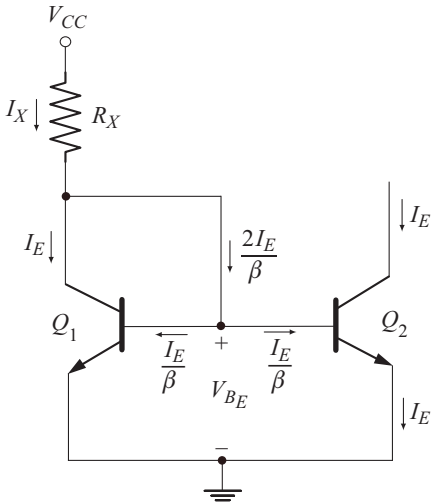


Fig. 5.16 Current mirror circuit

Mathematical Analysis

The base currents of Q_1 and Q_2 are given by

$$I_B = \frac{I_E}{1 + \beta}$$

$$\approx \frac{I_E}{\beta}$$

The collector current of each transistor is

$$I_C \approx I_E$$

The current I_x , through the resistor R_x is

$$\begin{aligned}
 I_X &= I_E + \frac{2I_E}{\beta} \\
 &= \left[1 + \frac{2}{\beta} \right] I_E \\
 &= \frac{\beta + 2}{\beta} I_E \\
 I_X &\approx I_E
 \end{aligned} \tag{5.45}$$

$$I_X = \frac{V_{CC} - V_{BE}}{R_X} \approx \frac{V_{CC}}{R_X} = \text{constant} \quad (5.46)$$

[Since $V_{BE} \ll V_{CC}$]

Note that I_x is mirrored in the collector of Q_2 .

5.9.1 Current Mirror with High Output Impedance

Figure 5.17 shows the circuit of current mirror with high output impedance. Transistor Q_2 is diode connected. The base emitter voltage is the same for both Q_1 and Q_2 .

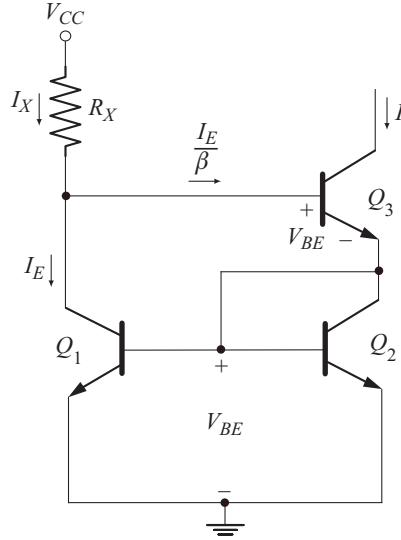


Fig. 5.17 Current mirror circuit with higher output impedance

$$I_X = \frac{V_{CC} - 2V_{BE}}{R_X} \approx \frac{V_{CC}}{R_X} = \text{constant} \quad (5.47)$$

If Q_1 and Q_2 are well matched, their collector currents are equal.

$$\therefore I \approx I_E \quad (5.47)$$

Now the base current of Q_3 is approximately $\frac{I_E}{\beta}$.

Summing the currents at the collector node of Q_1

$$I_X = I_E + \frac{I_E}{\beta} \approx I_E$$

$$\text{We find that } I \approx I_E \approx I_X \approx \frac{V_{CC}}{R_X} \quad (5.48)$$

Note that the current I is a mirrored value of the constant current I_X .

5.9.2 Current Mirror using JFET

Figure 5.18 shows the current mirror circuit using junction field effect transistor (JFET). JFET operates in the saturation region providing a constant current I_{DSS} since its gate and drain terminals are tied together.

The collector current of diode connected transistor Q_1 is I_{DSS} . Q_1 and Q_2 are driven by the same base-emitter voltages. If the two transistors are identical, they carry the same collector current. Thus I equals I_{DSS} .

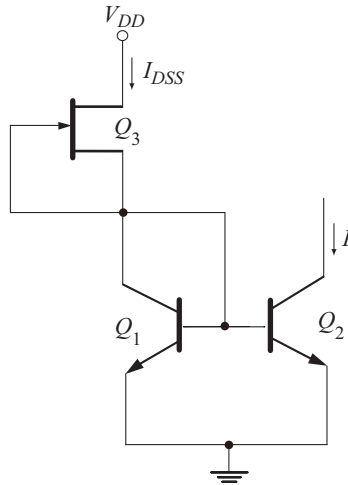
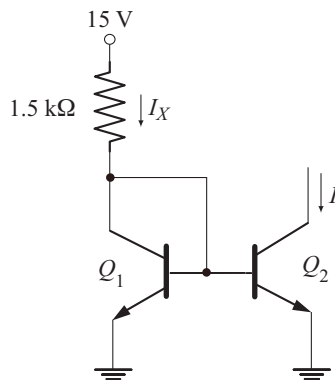


Fig. 5.18 Current mirror using JFET

Example 5.7

Calculate the mirrored current I in the circuit shown below. Take $\beta = 200$ for both the transistors.

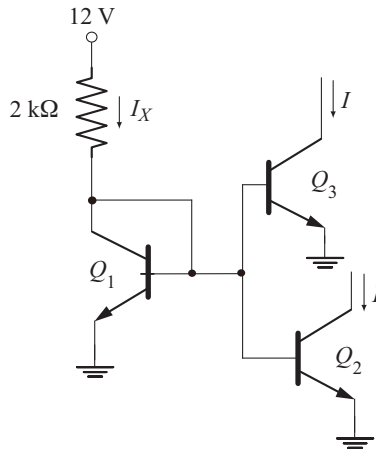


Solution

$$\begin{aligned}
 I = I_X &= \frac{V_{CC} - V_{BE}}{R_X} \\
 &= \frac{15\text{V} - 0.7\text{V}}{1.5\text{k}\Omega} \\
 &= 9.53\text{ mA}
 \end{aligned}$$

Example 5.8

Calculate the current I through each of the transistor Q_2 and Q_3 shown below.

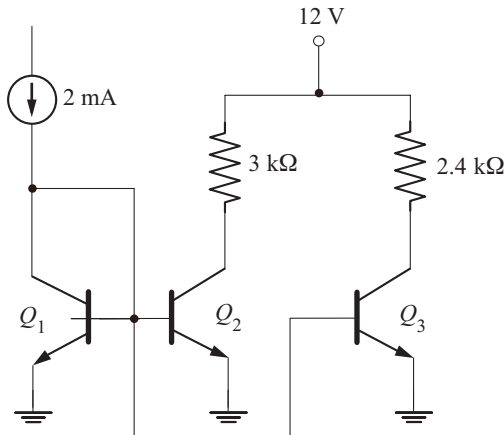
**Solution**

I_X equals the sum of collector current of Q_1 and the base currents of Q_1 , Q_2 and Q_3

$$\begin{aligned} \therefore I_X &= I_E + 3 I_B \\ &= I_E + 3 \frac{I_E}{\beta} = I_E \left[\frac{\beta + 3}{\beta} \right] \\ I_X &\approx I_E \approx I \\ I_X = I &= \frac{V_{CC} - V_{BE}}{R_X} = \frac{12\text{V} - 0.7\text{V}}{2\text{k}\Omega} = 5.65\text{ mA} \end{aligned}$$

Example 5.9

Calculate the collector currents of Q_2 and Q_3 . Take $\beta = 250$ for all transistors.



Solution

The collector current of Q_1 is 2 mA. Since the base-emitter voltages of Q_1 , Q_2 and Q_3 are same, a constant current of 2 mA is mirrored in the collectors of Q_2 and Q_3 .

◆ 5.10 CURRENT SOURCE CIRCUITS

A practical current source is represented by a source current in parallel with its source resistance. An ideal current source has infinite source resistance. A good practical current source has large source resistance. Figure 5.19 (a) and 5.19 (b) shows the representation of practical and ideal current sources.

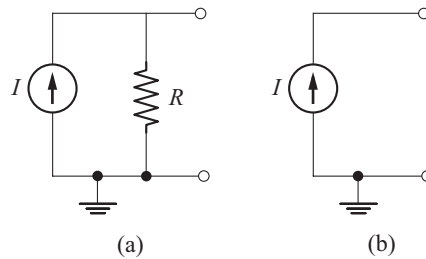


Fig. 5.19 (a) Practical current source (b) Ideal current source

An ideal current source provides a constant current independent of the load connected to it. Constant current sources with very high output resistance finds numerous applications in electronics. Constant-current circuits can be built using BJTs, FETs and a combination of these devices.

◆ 5.11 BJT CONSTANT-CURRENT SOURCE

Figure 5.20 shows a constant-current source using *nnpn* transistor. The combination of R_1 and R_2 forms a voltage divider between $-V_{EE}$ and ground.

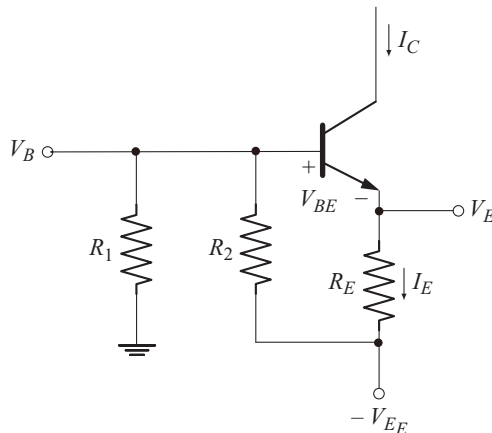


Fig. 5.20 BJT constant-current source

Neglecting base current, the base voltage with respect to ground, using voltage division rule is given by

$$V_B = (-V_{EE}) \frac{R_1}{R_1 + R_2} \quad (5.49)$$

$$\Rightarrow \begin{aligned} V_{BE} &= V_B - V_E \\ V_E &= V_B - V_{BE} \end{aligned} \quad (5.50)$$

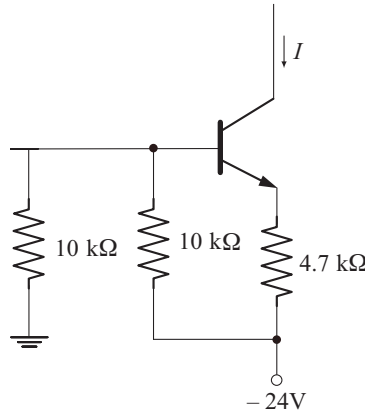
$$I_E = \frac{V_E - (-V_{EE})}{R_E} = \frac{V_E + V_{EE}}{R_E} \approx I_C \quad (5.51)$$

I_C is the constant current provided by the circuit of Fig. 5.20.

The main draw back of this circuit is that I_C depends on V_{BE} (since V_E contains V_{BE} term) which is temperature dependent. Thus I_C varies with temperature. An improved constant current source which uses zener diode in place of R_2 is discussed in the next section.

Example 5.10

Calculate the constant current I in the circuit shown below. Take $\beta = 100$.



Solution

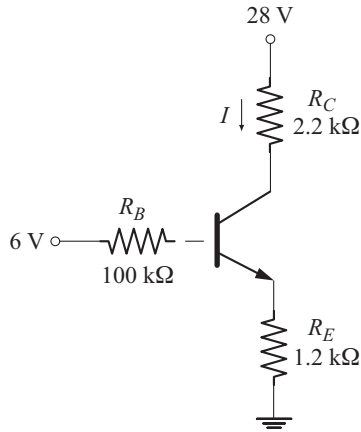
$$\begin{aligned} V_B &= (-V_{EE}) \frac{R_1}{R_1 + R_2} \\ &= (-24 \text{ V}) \frac{10 \text{ k}\Omega}{10 \text{ k}\Omega + 10 \text{ k}\Omega} = -12 \text{ V} \end{aligned}$$

$$V_E = V_B - 0.7 \text{ V} = -12 \text{ V} - 0.7 \text{ V} = -12.7 \text{ V}$$

$$\begin{aligned} I &= I_E = \frac{V_E - (-V_{EE})}{R_E} \\ &= \frac{-12.7 \text{ V} + 24 \text{ V}}{4.7 \text{ k}\Omega} = 2.4 \text{ mA} \end{aligned}$$

Example 5.11

Calculate the current I in the circuit shown below. Take $\beta = 120$.

**Solution**

$$I_B = \frac{V_B - V_{BE}}{R_B + (1 + \beta)R_E} = \frac{6\text{V} - 0.7\text{V}}{100\text{k}\Omega + (121)(1.2\text{k}\Omega)} = 21.62\text{ }\mu\text{A}$$

$$I = I_C = \beta I_B = (120)(21.62\text{ }\mu\text{A}) = 2.59\text{ mA}$$

◆ 5.12 BJT CONSTANT-CURRENT SOURCE USING ZENER DIODE

An improved BJT constant current source using zener diode is shown in Fig. 5.21.

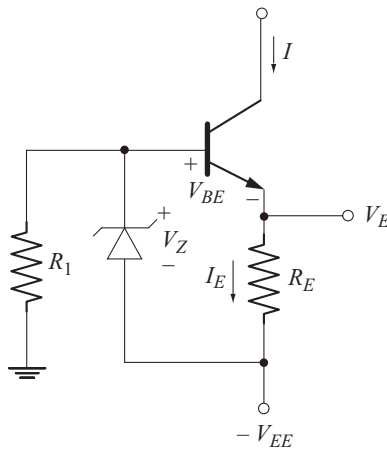


Fig. 5.21 Constant-current source using zener diode

Applying KVL to the base-emitter circuit we have

$$V_Z - V_{BE} - I_E R_E = 0$$

$$I_E = \frac{V_Z - V_{BE}}{R_E} \approx I \quad (5.52)$$

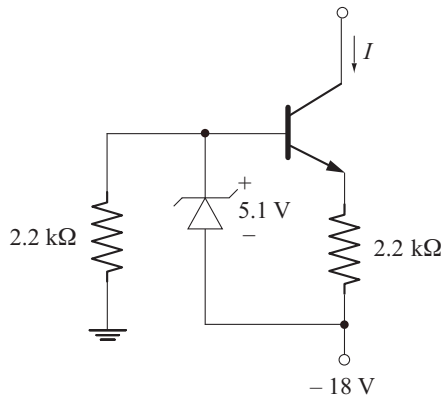
Usually $V_Z \gg V_{BE}$

$$\therefore I \approx I_E = \frac{V_Z}{R_E} \quad (5.53)$$

Note that the current I depends on V_Z and R_E which are both constants. Hence I is essentially a constant current. Also the current I is independent of the supply voltage V_{EE} .

Example 5.12

Calculate the constant current I in the circuit shown below. Take $\beta = 200$.



Solution

$$I \approx I_E = \frac{V_Z - V_{BE}}{R_E} = \frac{5.1\text{V} - 0.7\text{V}}{2.2\text{k}\Omega} = 2\text{ mA}$$

◆ 5.13 ANALYSIS OF TRANSISTOR CONFIGURATIONS USING APPROXIMATE HYBRID MODEL

In Chapter 3 we have analysed the various transistor configurations using the r_e model. Now let us proceed to analyse the same using the approximate hybrid model. Figure 5.22 shows the approximate CE hybrid model and the approximate CB hybrid model is shown in Fig. 5.23.

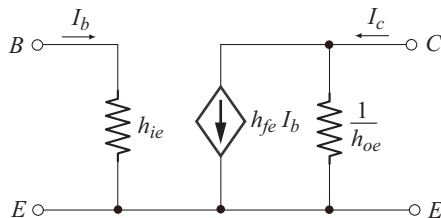


Fig. 5.22 Approximate CE hybrid model

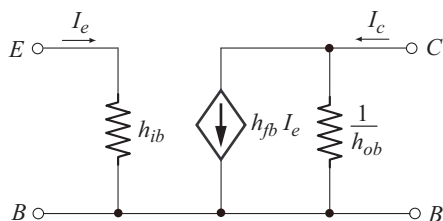


Fig. 5.23 Approximate CB hybrid model

Before starting the analysis, let us recall the relation between the parameters of r_e model and the hybrid model, which has been discussed in Chapter 3.

$$h_{ie} = \beta r_e \quad h_{fe} = \beta \quad h_{oe} = \frac{1}{r_o} \quad [\text{For CE configuration}]$$

$$h_{fb} = -\alpha \quad h_{ib} = r_e \quad [\text{For CB configuration}]$$

The approximate hybrid model is very similar in structure to that used with the r_e model. Therefore, all the results derived for the transistor configurations in Chapter 3 using r_e model can be readily applied for the analysis using the hybrid model by simply replacing the parameters of r_e model with their equivalent hybrid parameters.

◆ 5.14 COMMON-EMITTER FIXED BIAS CONFIGURATION

Figure 5.24 shows the circuit of common-emitter configuration using fixed bias. Its small signal ac equivalent circuit using approximate hybrid model is shown in Fig. 5.25.

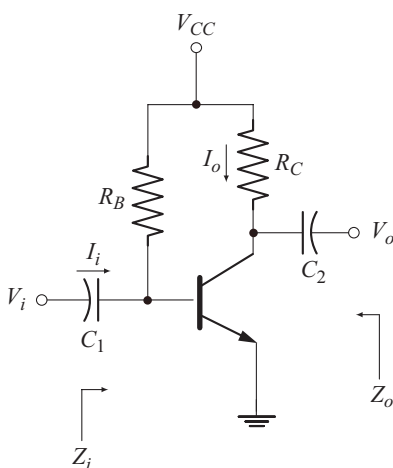


Fig. 5.24 CE fixed bias configuration

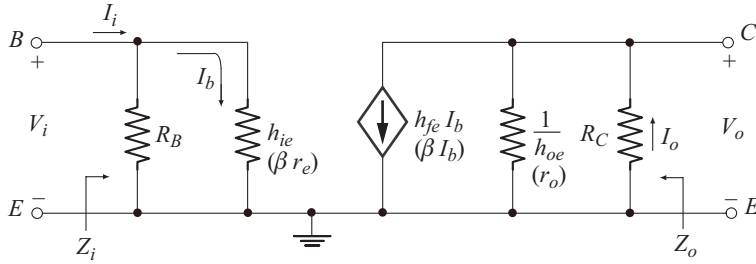


Fig. 5.25 AC equivalent circuit

The ac equivalent circuit of Fig. 5.25 is an exact replica of the circuit given in Fig. 3.36 in section 3.18 of Chapter 3. Applying the same procedure we obtain the following results.

- $$Z_i = R_B \parallel h_{ie} = R_B \parallel \beta r_e \quad (5.54)$$

- $$Z_i \approx h_{ie} = \beta r_e \quad [\text{If } R_B \gg h_{ie}] \quad (5.55)$$

- $$Z_o = \frac{1}{h_{oe}} \parallel R_C = r_o \parallel R_C \quad (5.56)$$

- $$A_V = - \frac{h_{fe} [r_o \parallel R_C]}{h_{ie}}$$

$$= - \frac{r_o \parallel R_C}{r_e} \quad (5.57)$$

- $$A_I = -A_V \frac{Z_i}{R_C} \quad (5.58)$$

For $\frac{1}{h_{oe}} \geq 10 R_C \equiv r_o \geq 10 R_C$

- $$Z_o \approx R_C \quad (5.59)$$

- $$A_V \approx - \frac{h_{fe} R_C}{h_{ie}} = - \frac{R_C}{r_e} \quad (5.60)$$

- $$A_I \approx -h_{fe} = -\beta \quad [\text{Taking } Z_i \approx h_{ie} = \beta r_e] \quad (5.61)$$

◆ 5.15 COMMON-EMITTER CONFIGURATION USING VOLTAGE DIVIDER BIAS

Figure 5.26 shows common-emitter configuration using voltage divider bias. The results of previous section can also be applied to this circuit. The only change is that R_B has to be replaced by $R_1 \parallel R_2$.

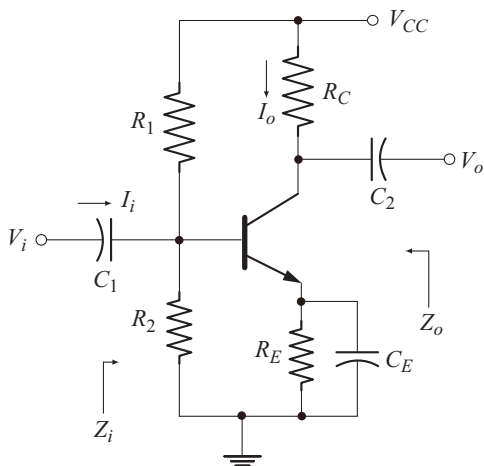


Fig. 5.26 CE Voltage divider bias configuration

Example 5.13

The following data are available for the amplifier circuit of Fig. 5.26.

$$\begin{array}{llll} R_1 = 68 \text{ k}\Omega & R_2 = 12 \text{ k}\Omega & R_C = 2.2 \text{ k}\Omega & R_E = 1.2 \text{ k}\Omega \\ V_{CC} = 18 \text{ V} & h_{fe} = 180 & h_{ie} = 2.75 \text{ k}\Omega & h_{oe} = 25 \text{ }\mu\text{S} \end{array}$$

- Calculate Z_i and Z_o
- Calculate A_v and A_i
- Determine r_e and compare βr_e with h_{ie} .

Solution

$$\begin{aligned} \text{(a)} \quad Z_i &= R_1 \parallel R_2 \parallel h_{ie} \\ R_1 \parallel R_2 &= R' = 68 \text{ k}\Omega \parallel 12 \text{ k}\Omega = 10.2 \text{ k}\Omega \\ Z_i &= 10.2 \text{ k}\Omega \parallel 2.75 \text{ k}\Omega = 2.17 \text{ k}\Omega \\ Z_o &= \frac{1}{h_{oe}} \parallel R_C = R'_C \\ \frac{1}{h_{oe}} &= \frac{1}{25 \text{ }\mu\text{S}} = 40 \text{ k}\Omega \\ Z_o &= 40 \text{ k}\Omega \parallel 2.2 \text{ k}\Omega = 2.09 \text{ k}\Omega \\ \text{(b)} \quad A_v &= - \frac{h_{fe} R'_C}{h_{ie}} \\ &= - \frac{(180)(2.09 \text{ k}\Omega)}{2.75 \text{ k}\Omega} = -136.8 \end{aligned}$$

$$A_I = - \frac{A_V Z_i}{R_C} = - \frac{(-136.8)(2.17 \text{ k}\Omega)}{2.2 \text{ k}\Omega}$$

$$= 134.93$$

(c) Check for $\beta R_E \geq 10 R_2$

$$\beta = h_{fe} = 180$$

$$\beta R_E = (180)(1.2 \text{ k}\Omega) = 216 \text{ k}\Omega$$

$$10 R_2 = (10)(12 \text{ k}\Omega) = 120 \text{ k}\Omega$$

Since $\beta R_E > 10 R_2$, we can use approximate analysis.

$$V_B = \frac{V_{CC} R_2}{R_1 + R_2} = \frac{18 \text{ V}(12 \text{ k}\Omega)}{68 \text{ k}\Omega + 12 \text{ k}\Omega} = 2.7 \text{ V}$$

$$V_E = V_B - V_{BE} = 2.7 \text{ V} - 0.7 \text{ V} = 2 \text{ V}$$

$$I_E = \frac{V_E}{R_E} = \frac{2 \text{ V}}{1.2 \text{ k}\Omega} = 1.67 \text{ mA}$$

$$r_e = \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{1.67 \text{ mA}} = 15.56 \Omega$$

$$\beta r_e = (180)(15.56 \Omega) = 2.8 \text{ k}\Omega$$

Note that βr_e is nearly equal to h_{ie} .

◆ 5.16 CE-EMITTER BIAS CONFIGURATION WITH UNBYPASSED R_E

Figure 5.27 shows the CE emitter bias configuration with unbypassed R_E . The small signal ac equivalent circuit using approximate hybrid model is shown in Fig. 5.28. To simplify the analysis

$\frac{1}{h_{oe}}$ is treated as an open circuit.

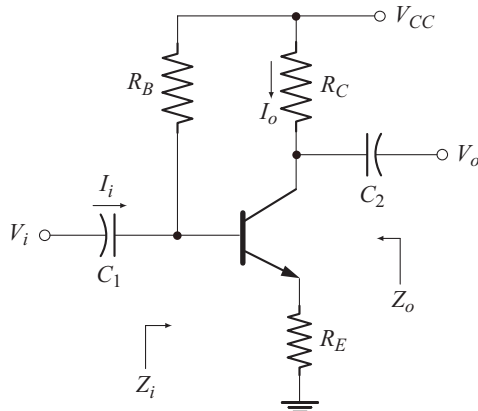


Fig. 5.27 Emitter bias configuration with un bypassed R_E

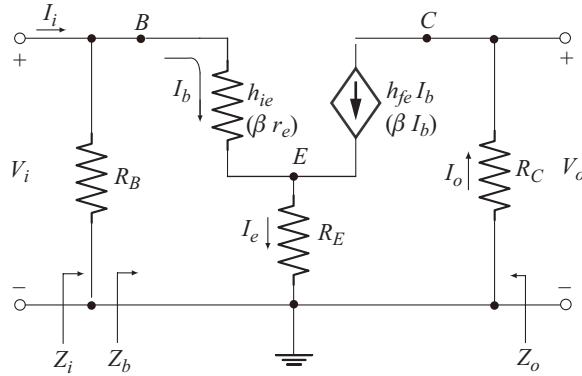


Fig. 5.28 Small signal ac equivalent circuit

The ac equivalent circuit of Fig. 5.28 is an exact duplicate of the circuit given in Fig. 3.45 in section 3.20 of Chapter 3. Applying the same procedure we obtain the following results.

- $$Z_b = h_{ie} + (1 + h_{fe}) R_E = \beta r_e + (1 + \beta) R_E \quad (5.62)$$

- $$Z_i \approx R_B \parallel Z_b \quad (5.63)$$

- $$A_V = - \frac{h_{fe} R_C}{h_{ie} + (1 + h_{fe}) R_E} \quad (5.64)$$

- $$A_I = - A_V \frac{Z_i}{R_C} \quad (5.65)$$

- $$Z_o = R_C \quad (5.66)$$

Using the approximations

$$1 + h_{fe} \approx h_{fe} \equiv 1 + \beta \approx \beta$$

$$h_{fe} R_E \gg h_{ie} \equiv \beta R_E \gg \beta r_e \quad \text{we get}$$

- $$Z_b \approx h_{fe} R_E = \beta R_E \quad (5.67)$$

- $$A_V \approx - \frac{R_C}{R_E} \quad (5.68)$$

Example 5.14

For the amplifier circuit of Fig. 5.27 the following data are available.

$$\begin{array}{llll} R_B = 330 \text{ k}\Omega & R_E = 1 \text{ k}\Omega & R_C = 3.3 \text{ k}\Omega & V_{CC} = 12 \text{ V} \\ h_{fe} = 120 & h_{ie} = 2 \text{ k}\Omega & h_{oe} = 20 \text{ mA/V} & \end{array}$$

- (a) Calculate Z_i and Z_o
- (b) Calculate A_V and A_I
- (c) Determine V_o if $V_i = 100$ mV

Solution

- (a)
- $$Z_b = h_{ie} + (1 + h_{fe}) R_E$$
- $$= 2 \text{ k}\Omega + (121)(1 \text{ k}\Omega) = 123 \text{ k}\Omega$$
- $$Z_i = R_B \parallel Z_b = 330 \text{ k}\Omega \parallel 123 \text{ k}\Omega = 89.6 \text{ k}\Omega$$
- $$Z_o = R_C = 3.3 \text{ k}\Omega$$
- (b)
- $$A_V = - \frac{h_{fe} R_C}{h_{ie} + (1 + h_{fe}) R_E}$$
- $$= - \frac{(120)(3.3 \text{ k}\Omega)}{2 \text{ k}\Omega + (121)(1 \text{ k}\Omega)} = -3.22$$
- $$A_I = - \frac{A_V Z_i}{R_C} = - \frac{(-3.22)(89.6 \text{ k}\Omega)}{3.3 \text{ k}\Omega} = 87.43$$
- (c)
- $$V_o = A_V V_i = (-3.22)(100 \text{ mV}) = -322 \text{ mV}$$

◆ 5.17 EMITTER FOLLOWER CONFIGURATION

Figure 5.29 shows the circuit of emitter follower. The small signal ac equivalent circuit of emitter follower using approximate hybrid model is shown in Fig. 5.30. For simplicity h_{oe} is not included in the model.

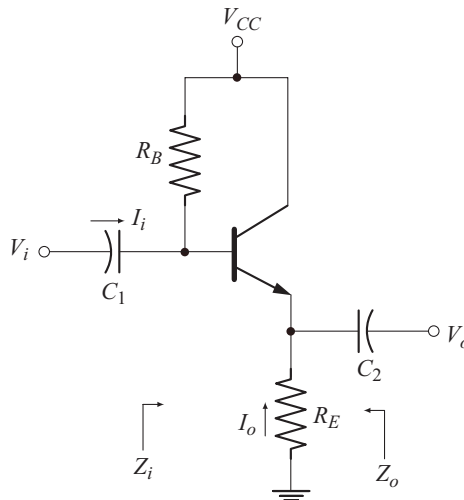


Fig. 5.29 Emitter-follower configuration

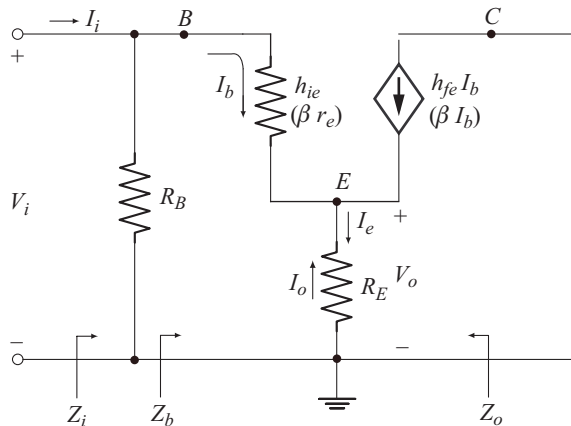


Fig. 5.30 Small signal ac equivalent circuit of emitter follower

- $$Z_b = h_{ie} + (1 + h_{fe}) R_E = \beta r_e + (1 + \beta) R_E \quad (5.69)$$

- $$\approx h_{fe} R_E = \beta R_E \quad (5.70)$$

- $$Z_i = Z_b \parallel R_B \quad (5.71)$$

- $$Z_o = R_E \parallel \frac{h_{ie}}{1 + h_{fe}} = R_E \parallel r_e \quad (5.72)$$

$$\approx \frac{h_{ie}}{h_{fe}} = r_e \quad (5.73)$$

- $$A_V = \frac{(1 + h_{fe}) R_E}{h_{ie} + (1 + h_{fe}) R_E} \quad (5.74)$$

$$\approx \frac{h_{fe} R_E}{h_{ie} + h_{fe} R_E} \quad (5.75)$$

$$= \frac{R_E}{r_e + R_E} \approx 1 \quad (5.76)$$

- $$A_I \approx - \frac{A_V Z_i}{R_C} \quad (5.77)$$

◆ 5.18 COMMON-BASE CONFIGURATION

Figure 5.31 shows the circuit of common base configuration. Its small signal ac equivalent circuit using approximate hybrid model is shown in Fig. 5.32. Again for simplicity h_{ob} is not included in the model.

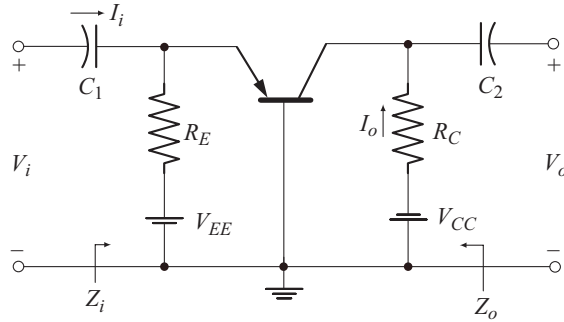


Fig. 5.31 Common-base configuration

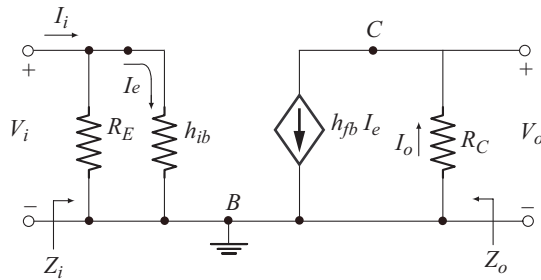


Fig. 5.32 Small signal ac equivalent circuit of CB configuration

Using the procedure given in section 3.27 of Chapter 3 we obtain the following results.

- $Z_i = R_E \parallel h_{ib} = R_E \parallel r_e \approx r_e$
- $Z_o = R_C$
- $A_V = -\frac{h_{fb} R_C}{h_{ib}} = \frac{\alpha R_C}{r_e}$
- $A_I = -h_{fb} = \alpha$

◆ 5.19 ANALYSIS OF GENERAL TRANSISTOR CONFIGURATION USING THE COMPLETE HYBRID EQUIVALENT MODEL

So far we have analyzed all the three transistor configurations by substituting the relevant approximate hybrid model in the ac equivalent circuit. In this section we are analyzing the transistor amplifier using the general complete hybrid equivalent model, which has all the four parameters h_i , h_f , h_r and h_o . The results can be modified for the specific transistor configuration by replacing the general hybrid parameters with the appropriate hybrid parameters of transistor configuration under consideration.

We begin our analysis by writing the transistor amplifier as a two port network, driven by a source V_s of internal resistance R_s and driving a load R_L as shown in Fig. 5.33.

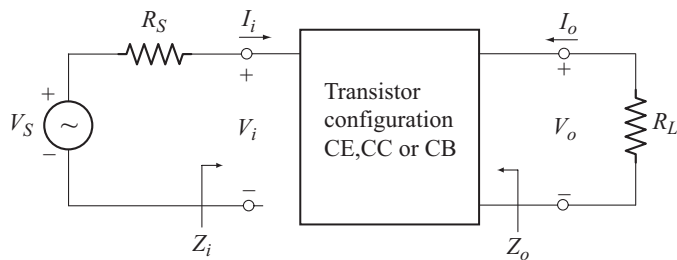


Fig. 5.33 Two-port system

Let us replace the transistor by its general, complete hybrid equivalent model as shown in Fig. 5.34.

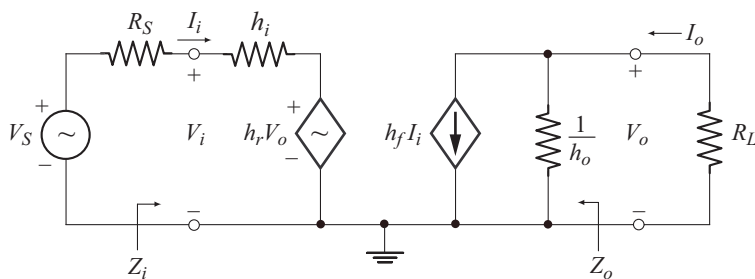


Fig. 5.34 Small signal ac equivalent circuit using the complete hybrid equivalent model

Current Gain [A_I]

The current gain is defined by

$$A_I = \frac{I_o}{I_i} \quad (5.78)$$

Applying KCL to the output circuit we have

$$\begin{aligned} I_o &= h_f I_i + h_o V_o \\ \text{Using } V_o &= -I_o R_L \text{ we get} \\ I_o &= h_f I_i + h_o (-I_o R_L) \\ I_o [1 + h_o R_L] &= h_f I_i \\ A_I &= \frac{I_o}{I_i} = \frac{h_f}{1 + h_o R_L} \end{aligned} \quad (5.79)$$

Input Impedance [Z_i]

Applying KVL to the input circuit of Fig. 5.34 we have

$$V_i - I_i h_i - h_r V_o = 0$$

$$V_i = I_i h_i + h_r [-I_o R_L] \quad (5.80)$$

But
$$A_I = \frac{I_o}{I_i} \Rightarrow I_o = A_I I_i$$

Using this relation in Equation (5.80) we get

$$\begin{aligned} V_i &= I_i h_i - h_r A_I R_L I_i \\ \text{Now } Z_i &= \frac{V_i}{I_i} = h_i - h_r A_I R_L \end{aligned} \quad (5.81)$$

Substituting for A_I from Equation (5.79) we have

$$Z_i = h_i - \frac{h_f h_r R_L}{1 + h_o R_L} \quad (5.82)$$

Voltage Gain [A_V]

$$\begin{aligned} V_o &= -I_o R_L \\ V_i &= I_i Z_i \\ \text{Now } A_V &= \frac{V_o}{V_i} = \frac{-I_o R_L}{I_i Z_i} \\ \text{or } A_V &= -\frac{A_I R_L}{Z_i} \end{aligned} \quad (5.83)$$

From Equation (5.82)

$$Z_i = \frac{h_i + (h_i h_o - h_f h_r) R_L}{1 + h_o R_L}$$

And from Equation (5.79)

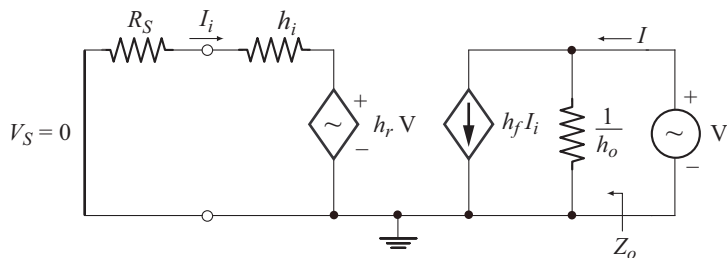
$$A_I = \frac{h_f}{1 + h_o R_L}$$

Using these relations in Equation (5.83) we obtain

$$\begin{aligned} A_V &= \frac{-h_f R_L}{1 + h_o R_L} \cdot \frac{1 + h_o R_L}{h_i + (h_i h_o - h_f h_r) R_L} \\ A_V &= \frac{-h_f R_L}{h_i + (h_i h_o - h_f h_r) R_L} \end{aligned} \quad (5.84)$$

Output Impedance [Z_o]

To find the output impedance, V_s is reduced to zero and a voltage source V is connected between the output terminals after removing R_L as shown in Fig. 5.35.

**Fig. 5.35** Circuit to find Z_o

Applying KCL to the output circuit we have

$$I = h_f I_i + h_o V \quad (5.85)$$

Writing KVL equation to the input circuit

$$\begin{aligned} -I_i(R_S + h_i) - h_r V &= 0 \\ I_i &= \frac{-h_r V}{R_S + h_i} \end{aligned}$$

Using this relation in Equation (5.85) we have

$$\begin{aligned} I &= -\frac{h_f h_r V}{R_S + h_i} + h_o V \\ \frac{I}{V} &= \frac{1}{Z_o} = h_o - \frac{h_f h_r}{R_S + h_i} \end{aligned} \quad (5.86)$$

$$Z_o = \frac{1}{h_o - [h_f h_r / (R_S + h_i)]} \quad (5.87)$$

$$\text{If } R_S = 0, \quad Z_o = \frac{1}{h_o - [h_f h_r / h_i]} \quad (5.88)$$

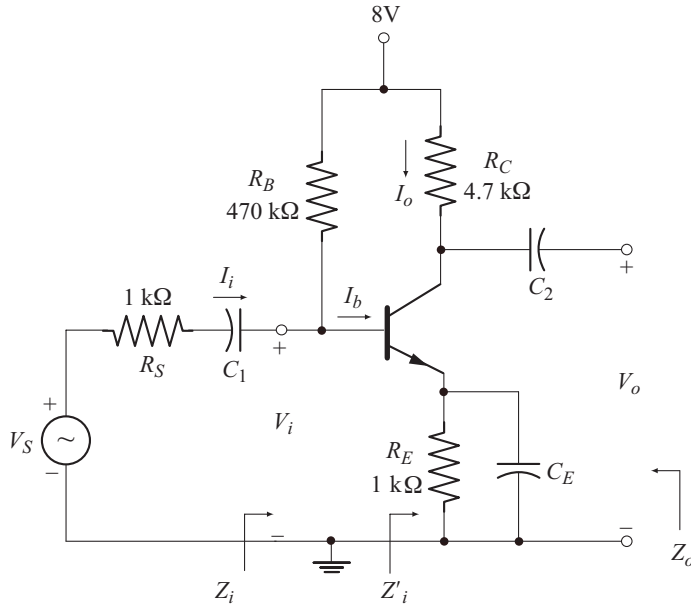
Example 5.15

For the circuit shown, using the complete hybrid equivalent model.

- Calculate Z'_i and Z_i
- Determine Z_o (including R_c) and Z'_o (without R_c)
- Calculate $A_I = \frac{I_o}{I_i}$ and $A'_I = \frac{I_o}{I_b}$
- Calculate A_V and A_{V_S}
- Repeat all calculations using approximate hybrid model and compare the results.

The hybrid parameters are

$$h_{ie} = 1.6 \text{ k}\Omega \quad h_{fe} = 110 \quad h_{re} = 2 \times 10^{-4} \quad h_{oe} = 20 \text{ }\mu\text{A/V}$$



Solution

The transistor is in CE configuration. Therefore in the general hybrid model of Fig. 5.40, we have to replace h_i , h_f , h_r and h_o by h_{ie} , h_{fe} , h_{re} and h_{oe} respectively.

The small signal ac equivalent circuit using the complete hybrid model is shown in Fig. A.

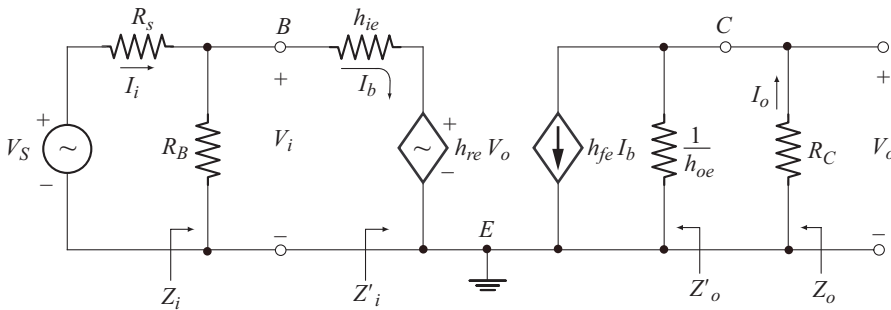


Fig. A

To obtain the correct results, it is important to compare the circuits of Fig. 5.34 and Fig. A and appropriately use the results obtained in section 5.19.

(a) Input Impedance [Z_i]

From Equation (5.82)

$$Z_i = h_{ie} - \frac{h_{fe} h_{re} R_L}{1 + h_{oe} R_L}$$

Derivation of Z_i does not include R_B . Hence in the present context the input impedance given by the above equation is Z_i' as indicated in Fig. A.

$$\begin{aligned} \text{Also } R_L &= R_C \\ \therefore Z_i' &= \frac{V_i}{I_b} = h_{ie} - \frac{h_{fe} h_{re} R_C}{1 + h_{oe} R_C} \quad (A) \\ &= 1.6 \text{ k}\Omega - \frac{(110)(2 \times 10^{-4})(4.7 \text{ k}\Omega)}{1 + (20 \times 10^{-6})(4.7 \text{ k}\Omega)} = 1.51 \text{ k}\Omega \end{aligned}$$

Input impedance taking R_B into account is

$$Z_i = \frac{V_i}{I_i} = R_B \parallel Z_i' = 470 \text{ k}\Omega \parallel 1.51 \text{ k}\Omega = 1.51 \text{ k}\Omega.$$

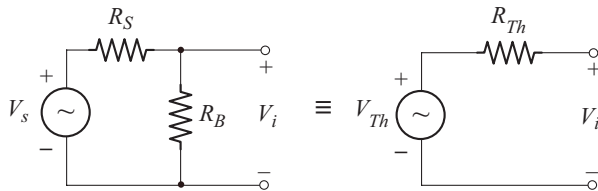
[For voltage divider bias, $R_B = R_1 \parallel R_2$]

(b) Output Impedance

From Equation (5.87)

$$Z_o = \frac{1}{h_{oe} - \left[\frac{h_{fe} h_{re}}{R_s + h_{ie}} \right]}$$

In the present situation the output impedance given by the above equation is Z_o' . Also R_s should be replaced by the Thevenin resistance R_{Th} of the circuit comprising of V_s , R_s and R_B which is shown below.



$$R_{Th} = R_s \parallel R_B = 1 \text{ k}\Omega \parallel 470 \text{ k}\Omega = 0.998 \text{ k}\Omega \quad [\text{If } R_s = 0, R_{Th} = 0]$$

$$\begin{aligned} Z_o' &= \frac{1}{h_{oe} - \left[\frac{h_{fe} h_{re}}{R_{Th} + h_{ie}} \right]} \quad (B) \\ Z_o' &= \frac{1}{20 \times 10^{-6} - \left[\frac{(110)(2 \times 10^{-4})}{0.998 \text{ k}\Omega + 1.6 \text{ k}\Omega} \right]} = 86.72 \text{ k}\Omega \end{aligned}$$

The output impedance taking R_C into account is

$$Z_o = Z_o' \parallel R_C = 86.72 \text{ k}\Omega \parallel 4.7 \text{ k}\Omega = 4.46 \text{ k}\Omega$$

(c) Current Gain

From Equation (5.79)

$$A_I = \frac{I_o}{I_i} = \frac{h_{fe}}{1 + h_{oe} R_L}$$

In the present situation the current gain given by the above equation is $A_I' = \frac{I_o}{I_b}$. also $R_L = R_C$.

$$\begin{aligned} A_I' &= \frac{I_o}{I_b} = \frac{h_{fe}}{1 + h_{oe} R_C} \\ &= \frac{110}{1 + (20 \times 10^{-6})(4.7 \text{ k}\Omega)} = 100.55 \\ A_I &= \frac{I_o}{I_i} = \frac{I_o}{I_b} \cdot \frac{I_b}{I_i} = (100.55) \frac{I_b}{I_i} \end{aligned} \quad (C)$$

Applying current division rule in the input circuit of Fig. A we have

$$\begin{aligned} I_b &= \frac{I_i R_B}{R_B + Z_i'} \\ \frac{I_b}{I_i} &= \frac{R_B}{R_B + Z_i'} = \frac{470 \text{ k}\Omega}{470 \text{ k}\Omega + 1.51 \text{ k}\Omega} = 0.997 \\ \text{Now } A_I &= (100.55) (0.997) = 100.25 \end{aligned}$$

(e) Voltage Gain

$$\begin{aligned} A_V &= \frac{V_o}{V_i} = \frac{-I_o R_C}{I_i Z_i} = - \frac{A_I R_C}{Z_i} \\ &= - \frac{(100.25)(4.7 \text{ k}\Omega)}{1.51 \text{ k}\Omega} = -312 \end{aligned} \quad (D)$$

$$A_{V_S} = \frac{V_o}{V_S} = \frac{V_o}{V_i} \cdot \frac{V_i}{V_S} = A_V \frac{V_i}{V_S} \quad (E)$$

Using voltage division rule in the circuit of Fig (B) we have

$$\begin{aligned} \frac{V_i}{V_S} &= \frac{Z_i}{Z_i + R_S} \\ &= \frac{1.51 \text{ k}\Omega}{1.51 \text{ k}\Omega + 1 \text{ k}\Omega} \\ &= 0.602 \end{aligned}$$

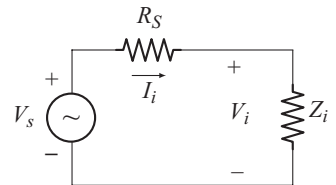


Fig B

Now $A_{V_S} = (-312)(0.602)$
 $= -187.82$ [If $R_S = 0$, $A_{V_S} = A_V$]

(e) To obtain the results using approximate hybrid model, we have to substitute $h_{re} = 0$ in Equations (A) to (E).

From Equation (A), with $h_{re} = 0$

$$\begin{aligned} Z'_i &= h_{ie} = 1.6 \text{ k}\Omega \\ Z_i &= R_B \parallel h_{ie} \\ &= 470 \text{ k}\Omega \parallel 1.6 \text{ k}\Omega = 1.595 \text{ k}\Omega \end{aligned}$$

From Equation (B)

$$\begin{aligned} Z'_o &= \frac{1}{h_{oe}} = \frac{1}{20 \times 10^{-6} \text{ S}} = 50 \text{ k}\Omega \\ Z_o &= Z'_o \parallel R_C = 50 \text{ k}\Omega \parallel 4.7 \text{ k}\Omega = 4.29 \text{ k}\Omega \\ A_I &= A_I \approx h_{fe} = 110 \\ A_V &= -\frac{A_I R_C}{Z_i} = -\frac{(110)(4.7 \text{ k}\Omega)}{1.595 \text{ k}\Omega} = -324.14 \\ A_{V_S} &= A_V \frac{Z_i}{Z_i + R_S} \\ &= (-324.14) \left[\frac{1.595 \text{ k}\Omega}{1.595 \text{ k}\Omega + 1 \text{ k}\Omega} \right] \\ &= -199.23 \end{aligned}$$

The following table compares the results obtained using exact and approximate hybrid models.

Parameter	Exact model	Approximate model
Z_i	1.51 k Ω	1.595 k Ω
Z_o	4.46 k Ω	4.29 k Ω
A_I	100.25	110
A_V	-312	-324.14

Note:

- A similar procedure can be used for CC [emitter follower] and CB configurations.
- If CE parameters are given, CB and CC parameters can be obtained using the conversion equations given in section 3.16.1 of Chapter 3.

Example 5.16

Given $\beta = 120$, $r_e = 4.5 \Omega$, $r_o = 40 \text{ k}\Omega$, $h_{re} = 2 \times 10^{-4}$

- Sketch the complete hybrid model
- Sketch the approximate hybrid model
- Sketch r_e model

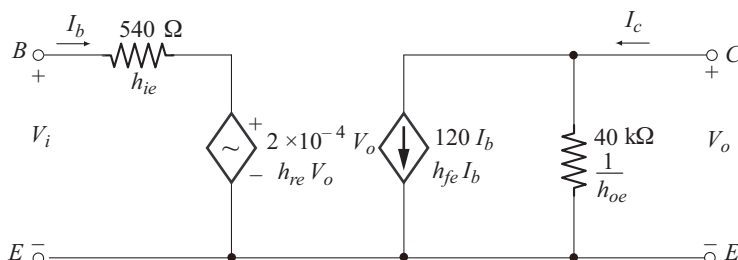
Solution

$$h_{ie} = \beta r_e = (120)(4.5 \Omega) = 540 \Omega$$

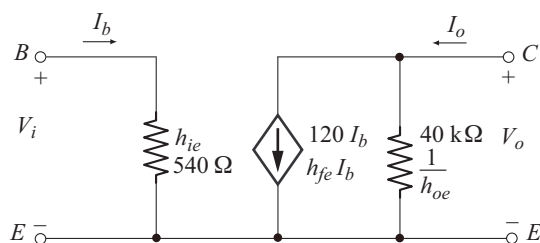
$$h_{fe} = \beta = 120$$

$$h_{oe} = \frac{1}{r_o} = \frac{1}{40 \text{ k}\Omega} = 25 \mu\text{S}$$

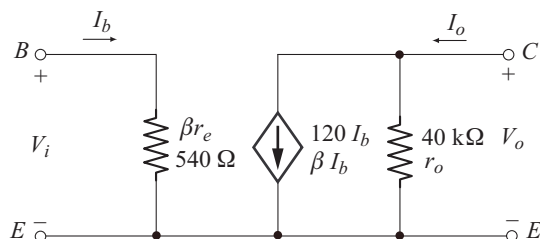
- (a) Complete hybrid model



- (b) Approximate hybrid model



- (c) r_e model



Note that the r_e model and the approximate hybrid model are identical except for the difference in symbols.

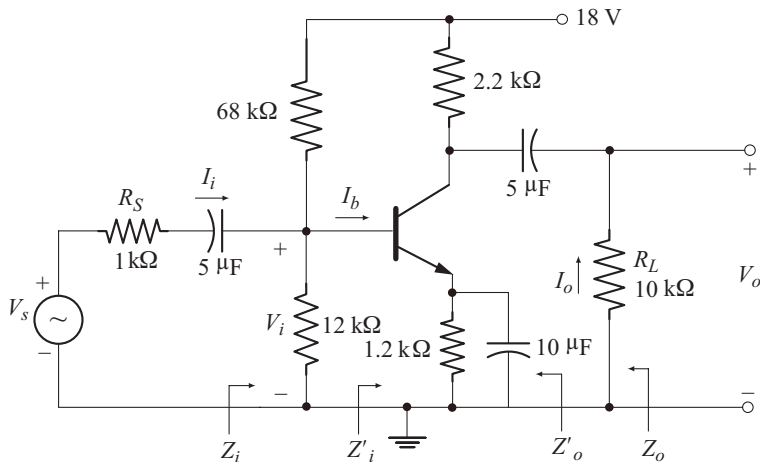
Example 5.17

For the circuit shown below, using the complete hybrid model:

- Calculate Z_i and Z'_i
- Calculate Z_o and Z'_o
- Determine $A_I = I_o/I_i$ and $A'_I = I_o/I_b$
- Determine A_V and A_{V_S}

The hybrid parameter values of the transistor are:

$$h_{fe} = 180 \quad h_{ie} = 2.75 \text{ k}\Omega \quad h_{oe} = 25 \text{ }\mu\text{S} \quad h_{re} = 2 \times 10^{-4}$$

**Solution**

The circuit contains both R_C and R_L . Hence it is better to derive the results and then calculate their numerical values. The small signal equivalent circuit using the complete hybrid model is shown in Fig. A.

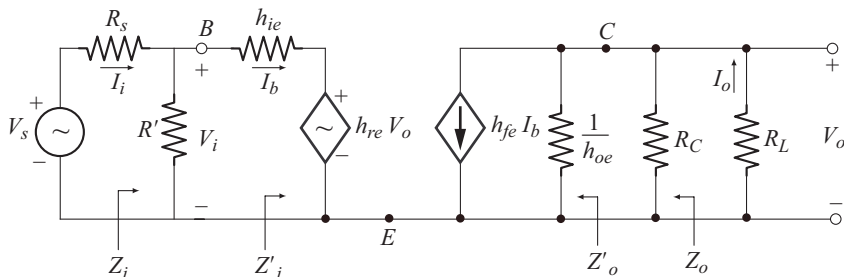


Fig. A

$$R' = R_1 \parallel R_2 = 68 \text{ k}\Omega \parallel 12 \text{ k}\Omega = 10.2 \text{ k}\Omega$$

(a) Input Impedance

Applying KVL to the input circuit we have

$$V_i = h_{ie} I_b + h_{re} V_o \quad (A)$$

$$\begin{aligned} \text{Let } R &= \frac{1}{h_{oe}} \parallel R_C \parallel R_L \\ R &= 40 \text{ k}\Omega \parallel 2.2 \text{ k}\Omega \parallel 10 \text{ k}\Omega = 1.73 \text{ k}\Omega \end{aligned}$$

The simplified output circuit is shown in Fig. B.

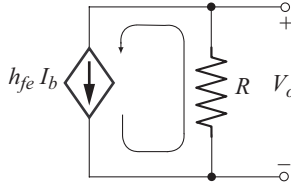


Fig B

From the circuit of Fig. B.

$$V_o = -h_{fe} I_b R$$

Using this relation in Equation (A) we have

$$\begin{aligned} V_i &= h_{ie} I_b - h_{fe} h_{re} R I_b \\ \text{Now } Z'_i &= \frac{V_i}{I_b} = h_{ie} - h_{fe} h_{re} R \quad (B) \\ &= 2.75 \text{ k}\Omega - (180)(2 \times 10^{-4})(1.73 \text{ k}\Omega) \\ &= 2.69 \text{ k}\Omega \end{aligned}$$

$$Z_i = \frac{V_i}{I_i} = R' \parallel Z'_i = 10.2 \text{ k}\Omega \parallel 2.69 \text{ k}\Omega = 2.13 \text{ k}\Omega$$

(b) Output Impedance

Z'_o does not include R_C . Hence

$$\begin{aligned} Z'_o &= \frac{1}{h_{oe} - \frac{h_{fe} h_{re}}{R_{Th} + h_{ie}}} \\ R_{Th} &= R' \parallel R_S = 10.2 \text{ k}\Omega \parallel 1 \text{ k}\Omega = 0.91 \text{ k}\Omega \\ Z'_o &= \frac{1}{25 \times 10^{-6} - \frac{(180)(2 \times 10^{-4})}{0.91 \text{ k}\Omega + 2.75 \text{ k}\Omega}} = 65.95 \text{ k}\Omega \\ Z_o &= Z'_o \parallel R_C = 65.95 \text{ k}\Omega \parallel 2.2 \text{ k}\Omega = 2.14 \text{ k}\Omega \end{aligned}$$

(c) Current Gain

Applying KCL to the output circuit of Fig. A we have

$$I_o = h_{fe} I_b + h_{oe} V_o + \frac{V_o}{R_C}$$

Using $V_o = -I_o R_L$, we get

$$I_o = h_{fe} I_b + h_{oe} (-I_o R_L) + \frac{(-I_o R_L)}{R_C}$$

$$I_o \left[1 + h_{oe} R_L + \frac{R_L}{R_C} \right] = h_{fe} I_b$$

Now

$$A'_I = \frac{I_o}{I_b} = \frac{h_{fe}}{1 + h_{oe} R_L + \frac{R_L}{R_C}} \quad (C)$$

$$= \frac{180}{1 + (25 \times 10^{-6})(10\text{k}\Omega) + \left[\frac{10\text{k}\Omega}{2.2\text{k}\Omega} \right]}$$

$$= 31.06$$

$$A_I = \frac{I_o}{I_i} = A'_I \cdot \frac{Z'_i}{R' + Z'_i}$$

$$= (31.06) \frac{2.69\text{k}\Omega}{10.2\text{k}\Omega + 2.69\text{k}\Omega}$$

$$= 6.48$$

(d) Voltage Gain

$$A_V = \frac{V_o}{V_i} = \frac{-I_o R_L}{I_i Z_i}$$

$$= -A_I \cdot \frac{R_L}{Z_i} = -(6.42) \frac{10\text{k}\Omega}{2.13\text{k}\Omega}$$

$$= -30.14$$

$$A_{Vs} = A_V \cdot \frac{Z_i}{Z_i + R_s}$$

$$= (-30.14) \frac{2.13\text{k}\Omega}{2.13\text{k}\Omega + 1\text{k}\Omega}$$

$$= -20.51$$

Note: The same procedure can be used to analyze CC and CB amplifiers with R_L .



Exercise Problems

- 5.1** A CE fixed bias amplifier with bypassed R_E has $R_C = 1.5 \text{ k}\Omega$, $R_B = 390 \text{ k}\Omega$, $R_E = 1.2 \text{ k}\Omega$, $h_{fe} = 120$, $h_{ie} = 2 \text{ k}\Omega$, $h_{oe} = 20 \text{ }\mu\text{S}$. Calculate Z_i , Z_o , A_V and A_I .
- 5.2** Repeat previous problem with unbypassed R_E .
- 5.3** An emitter follower using fixed bias has $R_B = 470 \text{ k}\Omega$, $R_E = 1 \text{ k}\Omega$, $h_{fe} = 120$, $h_{ie} = 2 \text{ k}\Omega$ and $h_{oe} = 20 \text{ }\mu\text{S}$. Calculate Z_i , Z_o , A_V and A_I .
- 5.4** A CB configuration has $R_E = 1.5 \text{ k}\Omega$, $R_C = 2.2 \text{ k}\Omega$, $h_{ib} = 10 \text{ k}\Omega$, $h_{fb} = -0.996$ and $h_{ob} = 0.4 \text{ }\mu\text{S}$. Calculate Z_i , Z_o , A_V and A_I .
- 5.5** Repeat Exercise 5.3, if $R_S = 1 \text{ k}\Omega$ and $R_L = 10 \text{ k}\Omega$.

Chapter 6

FEEDBACK AMPLIFIERS

Negative feedback is used to stabilise the amplifier against variations in component values and parameters of the active devices used in the circuit. Negative feedback reduces distortion, reduces noise output, improves frequency response and improves input and output resistances. All these advantages are obtained at the cost of transfer gain. This chapter discusses the effect of negative feedback on various amplifier parameters. Analysis of practical feedback amplifiers has also been discussed.

◆ 6.1 CLASSIFICATION OF AMPLIFIERS

Amplifiers can be classified based on the magnitude of the input and output impedances as follows:

1. Voltage amplifiers
2. Current amplifiers
3. Transconductance amplifiers
4. Transresistance amplifiers

6.1.1 Voltage Amplifier

As the name implies, a voltage amplifier provides an output voltage proportional to the input voltage. Ideally, the proportionality constant is independent of the source and the load resistances. The equivalent circuit of the voltage amplifier is shown in Fig. 6.1.

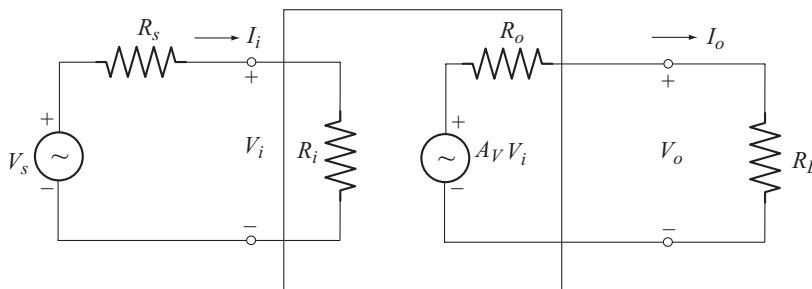


Fig. 6.1 Equivalent circuit of voltage amplifier

Observe in Fig. 6.1, that the voltage amplifier is driven by a voltage source V_s of internal resistance R_s . The output is represented by a voltage source $A_V V_i$ in series with output resistance R_o across which the load R_L is connected. R_i represents the input resistance of the amplifier.

Using voltage division rule at the input we have

$$V_i = \frac{V_s R_i}{R_s + R_i}$$

or

$$V_i = \frac{V_s}{1 + \left[\frac{R_s}{R_i} \right]} \quad (6.1)$$

It is desirable that the entire source voltage be available at the input terminals of the amplifier

$$\text{i.e.,} \quad V_i \approx V_s$$

From Equation (6.1) we find that, the above condition is satisfied if

$$\frac{R_s}{R_i} \ll 1$$

$$\Rightarrow R_i \gg R_s \quad (6.2)$$

i.e., The voltage amplifier must be designed with very large input resistance. Ideally $R_i = \infty$.

Using voltage division rule at the output we have

$$V_o = \frac{A_V V_i R_L}{R_o + R_L}$$

or

$$V_o = \frac{A_V V_i}{1 + \left[\frac{R_o}{R_L} \right]} \quad (6.3)$$

It is desirable that the entire amplified voltage be available across the load

$$\text{i.e.,} \quad V_o \approx A_V V_i \quad (6.4)$$

From Equation (6.3) we find that, the above condition is satisfied if

$$\frac{R_o}{R_L} \ll 1$$

$$\Rightarrow R_o \ll R_L \quad (6.5)$$

i.e., the voltage amplifier must be designed with very low output resistance. Ideally $R_o = 0$.

Now we conclude the following:

A good voltage amplifier must have a very large input resistance and a very low output resistance. Ideally $R_i = \infty$ and $R_o = 0$.

From Equation (6.4) we have

$$A_v = \frac{V_o}{V_i} \quad (6.6)$$

A_v is called the open loop voltage gain or voltage gain without feedback.

6.1.2 Current Amplifier

A current amplifier provides an output current proportional to the input current. Ideally, the proportionality constant is independent of the source and load resistances. The equivalent circuit of the current amplifier is shown in Fig. 6.2.

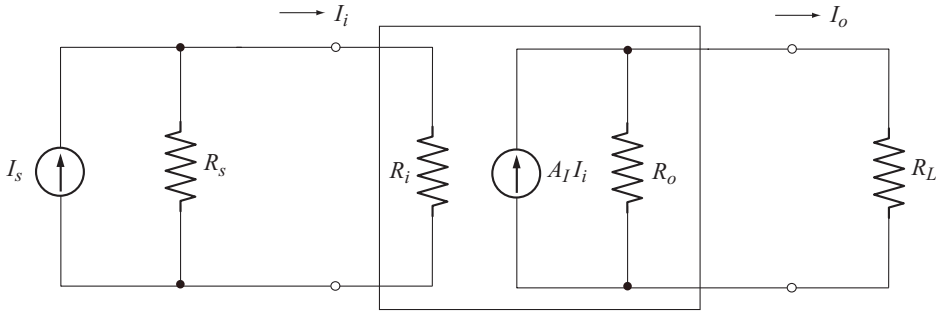


Fig. 6.2 Equivalent circuit of current amplifier

Observe that, the current amplifier is driven by a current source I_s of source resistance R_s . The output is also represented by a current source $A_i I_i$ in parallel with output resistance R_o . R_i represents the input resistance of the amplifier.

Using current division rule at the input we have

$$I_i = \frac{I_s R_s}{R_s + R_i}$$

$$I_i = \frac{I_s}{1 + \left[\frac{R_i}{R_s} \right]} \quad (6.7)$$

It is desirable to have the entire source current to flow into R_i

$$\text{i.e.,} \quad I_i \approx I_s$$

From Equation (6.7) we find that, the above condition is satisfied when

$$\frac{R_i}{R_s} \ll 1$$

$$\Rightarrow R_i \ll R_s \quad (6.8)$$

i.e., The current amplifier must be designed with very low input resistance. Ideally $R_i = 0$.

Using current division rule at the output we have

$$I_o = \frac{A_I I_i R_o}{R_o + R_L}$$

or

$$I_o = \frac{A_I I_i}{1 + \frac{R_L}{R_o}} \quad (6.9)$$

It is desirable to have the entire amplified current to flow into R_L .

$$\text{i.e.,} \quad I_o \approx A_I I_i \quad (6.10)$$

From Equation (6.9) we find that, the above requirement is satisfied, when

$$\frac{R_L}{R_o} \ll 1$$

$$\Rightarrow R_o \gg R_L \quad (6.11)$$

i.e., The current amplifier must be designed with very high output resistance. Ideally $R_o = \infty$.

Now we conclude the following:

A good current amplifier must have a very low input resistance and a very high output resistance. Ideally $R_i = 0$ and $R_o = \infty$.

From Equation (6.10) we have

$$A_I = \frac{I_o}{I_i} \quad (6.12)$$

A_I is called the open loop current gain or current gain without feedback.

6.1.3 Transconductance Amplifier

A transconductance amplifier provides an output current proportional to the input voltage. Ideally the proportionality constant is independent of the source and load resistances. Since, the output is a current and the input is a voltage, the proportionality constant has the unit of conductance and hence this arrangement is called a transconductance amplifier or voltage to current converter. The equivalent circuit of the transconductance amplifier is shown in Fig. 6.3.

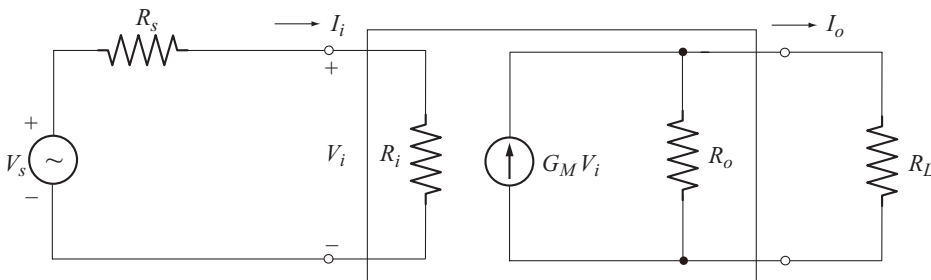


Fig. 6.3 Equivalent circuit of transconductance amplifier

Observe that, the amplifier is driven by a voltage source V_s of source resistance R_s . The output is represented by a current source $G_M V_i$ in parallel with output resistance R_o . R_i represents the input resistance of the amplifier.

Using voltage division rule at the input we have

$$V_i = \frac{V_s R_i}{R_s + R_i}$$

or

$$V_i = \frac{V_s}{1 + \left[\frac{R_s}{R_i} \right]} \quad (6.13)$$

It is desirable that the entire source voltage be available at the input terminals of the amplifier.

$$\text{i.e.,} \quad V_i \approx V_s$$

From Equation (6.13) we find that, the above requirement is met with if

$$\frac{R_s}{R_i} \ll 1$$

$$\Rightarrow R_i \gg R_s \quad (6.14)$$

i.e., The transconductance amplifier must be designed with very high input resistance. Ideally $R_i = \infty$.

Using current division rule at the output we have

$$I_o = \frac{G_M V_i R_o}{R_o + R_L}$$

or

$$I_o = \frac{G_M V_i}{1 + \left[\frac{R_L}{R_o} \right]} \quad (6.15)$$

It is desirable that, the entire output current $G_M V_i$ to flow into R_L .

$$\text{i.e.,} \quad I_o \approx G_M V_i \quad (6.16)$$

From Equation (6.15) we find that, the above requirement is satisfied when

$$\frac{R_L}{R_o} \ll 1$$

$$\Rightarrow R_o \gg R_L \quad (6.17)$$

i.e., The transconductance amplifier must be designed with very high output resistance. Ideally $R_o = \infty$.

Now we conclude the following:

A good transconductance amplifier must have very high input and output resistances. Ideally $R_i = \infty$ and $R_o = \infty$.

From Equation (6.16) we have

$$G_M = \frac{I_o}{V_i} \quad (6.18)$$

G_M is called the open loop transconductance or the transconductance without feedback.

6.1.4 Transresistance Amplifier

A transresistance amplifier provides an output voltage proportional to the input current. Ideally the proportionality constant is independent of the source and load resistances. Since the output is a voltage and the input is a current, the proportionality constant has the unit of resistance and hence this arrangement is called a transresistance amplifier or current to voltage converter. The equivalent circuit of the transresistance amplifier is shown in Fig. 6.4.

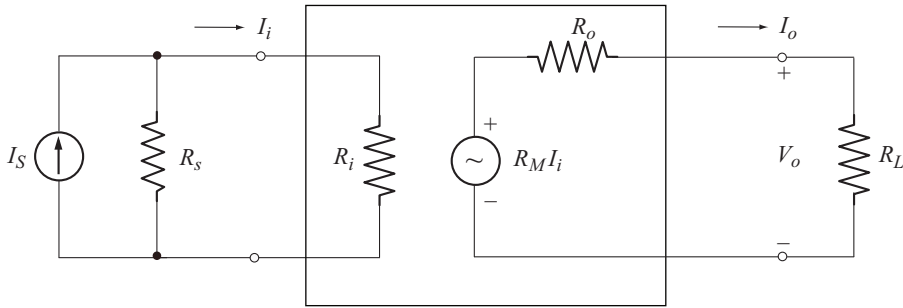


Fig. 6.4 Equivalent circuit of transresistance amplifier

Observe that, the amplifier is driven by a current source I_s of source resistance R_s . The output is represented by a voltage source $R_M I_i$ in series with output resistance R_o . R_i represents the input resistance of the amplifier.

Using current division rule at the input we have

$$I_i = \frac{I_s R_s}{R_s + R_i}$$

or

$$I_i = \frac{I_s}{1 + \frac{R_i}{R_s}} \quad (6.19)$$

It is desirable to have the entire source current to flow into R_i .

$$\text{i.e.,} \quad I_i \approx I_s$$

From Equation (6.19) we find that, the above requirement is met with when

$$\frac{R_i}{R_s} \ll 1$$

$$\Rightarrow R_i \ll R_s \quad (6.20)$$